The transverse scale of small-parameter exponential geodesics

AI-assisted research manuscript

Abstract

We study the maximum Euclidean displacement of a unit geodesic for exponential metrics of the logarithmic Gaussian field on $\mathbb{R}^d$. In each fixed dimension $d \geq 2$, we obtain the scale $\gamma^{2/3}$, bounds for every fixed positive moment, and a small-height bound with exponent $d-1$. The rescaled heights are tight and bounded away from zero in probability; their subsequential laws have unbounded support and two-sided cubic exponential tail bounds. The centered paths are also tight at this scale, with vanishing longitudinal component. The upper bound couples a quantitative Gaussian expansion to a midpoint comparison, controlling the height and the distance exponent together. The lower bound compares conformally bent tubes through their Cameron–Martin likelihoods. In higher dimensions we take the earlier metric characterization and conformal-covariance theorem as inputs. No construction or uniqueness proof for the metric is repeated here.

Status. This is an AI-assisted proposed research manuscript. The higher-dimensional conclusions are conditional on the characterization in [U] and the conformal-covariance result in [C], with their precise scope stated below.

Contents

1 Statement and proof outline **2**

2 Gaussian calculus **6**

3 Uniform metric estimates **7**

4 The first-order remainder **11**

5 Gaussian line integrals **14**

6 Random endpoints **16**

7 Height and exponent bounds **18**

8 Path tightness **23**

9 The lower tail **26**

10 Completion of the main results **31**

1 Statement and proof outline

A small exponential perturbation of Euclidean geometry displaces its geodesics from straight segments. We study the largest displacement along the geodesic from 0 to e_1. The purpose of this paper is to estimate that displacement, not to construct or characterize the underlying metric.

Fix an integer $d \geq 2$. Let h be the whole-space logarithmic Gaussian field with covariance coefficient one. Write σ_d for uniform probability measure on S^{d-1}, and pin the field by $\langle h, \sigma_d \rangle = 0$. Before pinning, its covariance on test functions of integral zero is $-\log|x-y|$. In dimension two this is the whole-plane Gaussian free field. For $d > 2$ it is the logarithmic field, not the ordinary higher-dimensional Gaussian free field.

1.1 Imported metric results

In dimension $d \geq 3$, we use the assignment in [U, Theorem 1.2, p. 2]. The uniqueness of weak assignments up to a deterministic positive scalar is the input [U, Theorem 1.4, p. 3], and exact dilation covariance is [U, Corollary 10.2, p. 27]. Orthogonal covariance is supplied by [C, Lemma 1.2, p. 2]. These statements concern the same whole-space logarithmic field, covariance coefficient, subcritical range $Q_\xi > \sqrt{2d}$, and class of possibly dependent continuous field perturbations used here. In dimension two we use the LQG metric of [GM, GMc].

The cited weak uniqueness identifies any weak LFPP assignment with the chosen strong assignment up to a deterministic scalar, separately at each parameter. Such a scalar leaves both geodesic images and constant-speed parametrizations on $[0,1]$ unchanged. Thus no comparison of geodesics under a merely bi-Lipschitz change of metric is needed. The approximation schemes identified with this assignment are exactly those covered by [U, Corollary 10.4, p. 28]; we make no extension to other mollifiers.

For the analytic argument, record the following properties of D_h^ξ, $0 < \xi < \xi_0$. Statements about each fixed deterministic coordinate change or domain hold almost surely; no common null set for uncountably many such choices is required.

(i) *Length, topology and locality.* The assignment is a measurable function of the field, defined for a whole-space logarithmic field plus a possibly dependent continuous function. It is a finite length metric inducing the Euclidean topology. For every deterministic open set U, its internal metric $D_h^\xi(\cdot,\cdot;U)$ is determined by $h|_U$. The length, topology and locality assertions apply to this full input class.

(ii) *Weyl scaling.* Almost surely, simultaneously for every continuous f, $D_{h+f}^\xi = e^{\xi f} \cdot D_h^\xi$, where the right side minimizes $e^{\xi f}$ against D_h^ξ-arclength.

(iii) *Exact similarities.* There are numbers $a_\xi = \xi Q_\xi$ with $a_\xi \to 1$ such that, for $z \in \mathbb{R}^d$, $r > 0$ and $O \in O(d)$,

$$D_h^\xi(z + rOx, z + rOy) = r^{a_\xi} e^{\xi h_r(z)} D_{h^{z,r,O}}^\xi(x,y), \qquad h^{z,r,O}(u) = h(z + rOu) - h_r(z)\,. \tag{1.1}$$

Here $h_r(z)$ is the spherical average and $h^{z,r,O}$ has the pinned law of the unperturbed Gaussian field h. Simultaneous Weyl scaling extends the pathwise coordinate identity to continuous perturbations.

(iv) *Fixed-parameter moments.* For each prescribed finite P, and every sufficiently small ξ,

$$\mathbb{E}\big[D_h^\xi(0,e_1)^P + D_h^\xi(0,e_1)^{-P}\big] < \infty\,. \tag{1.2}$$

No uniform bound for these moments is assumed.

The first three properties are the imported metric axioms, with orthogonal covariance as cited above. Property (iv) follows from the fixed-parameter moment estimates in [CHZ, Theorem 1.4(1)–(2), p. 5; Propositions 4.7 and 4.11], since the assignment is weak by [U, Lemma 10.1, pp. 26–27]. For positive moments, bound a point distance by the internal diameter of a compact connected set containing its endpoints. The admissible order is less than $2d/(\xi\gamma_\xi)$, where γ_ξ is the smaller positive solution of $Q_\xi = d/\gamma_\xi + \gamma_\xi/2$. This threshold tends to infinity as $\xi \downarrow 0$. For negative moments, use the distance between two disjoint non-singleton compact connected sets containing the two endpoints: that set distance is no larger than the point distance and has all negative moments. None of these quoted estimates needs a constant uniform in ξ.

For a Möbius map $\phi : V \to U$, smooth on a neighborhood of the domains in question, put $\psi = \phi^{-1}$ and define σ_ψ by $|D\psi(x)v| = \sigma_\psi(x)|v|$. The lower bound uses the following coordinate identity from [C, Theorem 1.1, p. 2] when $d \geq 3$, and from [GMc, Theorem 1.1] when $d = 2$:

$$D_h^\xi(x, y; V) = D^\xi_{h\circ\psi + Q_\xi \log \sigma_\psi}(\phi(x), \phi(y); U)\,. \tag{1.3}$$

The cited formula applies to the internal metrics and to continuous field perturbations, with a null event allowed to depend on the fixed map. This is the exact scope needed for the deterministic finite families of bends below. The fields and metrics are interpreted locally; no extension across a pole is used.

1.2 Parameters and conclusions

For the LFPP distance exponent, [DGZ, Proposition 1.1, p. 4] gives

$$\xi^{-1} - \sqrt{2d} \leq Q_\xi \leq \xi^{-1} + \sqrt{2}\,.$$

In particular $a_\xi = \xi Q_\xi \to 1$. Only this coarse convergence is imported; the sharper exponent bound in Theorem B is part of the argument below.

After decreasing ξ_0 so that $Q_\xi > \sqrt{2d}$, define γ_ξ as the smaller positive solution of

$$Q_\xi = \frac{d}{\gamma_\xi} + \frac{\gamma_\xi}{2}\,. \tag{1.4}$$

Since $a_\xi \to 1$, we have

$$\frac{\xi}{\gamma_\xi} = \frac{a_\xi}{d + \gamma_\xi^2/2} \longrightarrow \frac{1}{d}\,. \tag{1.5}$$

The proof is formulated for the ξ-indexed family. In the usual γ-parameterization, $D^\gamma = D^{\xi_\gamma}$ with $\gamma_{\xi_\gamma} = \gamma$ and $\xi_\gamma \to 0$. No injectivity or continuity of an arbitrary abstract function $\xi \mapsto a_\xi$ is assumed. Deterministic multiplicative normalizations of D_h^ξ do not affect its geodesics.

For the LFPP exponent, [DGZ, Proposition 1.1] also gives continuity of Q_ξ. Every sufficiently small conventional γ is therefore represented by some $\xi_\gamma > 0$ with $Q_{\xi_\gamma} = d/\gamma + \gamma/2$. Every such choice satisfies $\xi_\gamma \leq (Q_{\xi_\gamma} - \sqrt{2})^{-1} \to 0$, so (1.5) applies without requiring strict monotonicity.

Theorem A. *Under properties* (i)–(iv)*, for each fixed sufficiently small ξ, the* 0*-to-*e_1 *geodesic exists and is almost surely unique. Write P_ξ for its image and set*

$$M_\xi = \max_{z \in P_\xi} \operatorname{dist}(z, [0, e_1]), \qquad \kappa_d = d - 1\,.$$

For every fixed $p > 0$, there are constants $C_{d,p} < \infty$ and $\xi_{d,p} > 0$ such that

$$\mathbb{E}M_\xi^p \leq C_{d,p}\xi^{2p/3}, \qquad 0 < \xi < \xi_{d,p}\,. \tag{1.6}$$

There are also $C_d < \infty$ and $\xi_d > 0$ such that

$$\mathbb{P}[M_\xi \le a\xi^{2/3}] \le C_d a^{\kappa_d}, \qquad 0 < a \le 1, \quad 0 < \xi < \xi_d\,. \tag{1.7}$$

In particular, for every $\varepsilon > 0$ there are $0 < a < A < \infty$ and $\xi_\varepsilon > 0$ such that

$$\mathbb{P}\left[a \le \frac{M_\xi}{\xi^{2/3}} \le A\right] \ge 1 - \varepsilon, \qquad 0 < \xi < \xi_\varepsilon\,. \tag{1.8}$$

For every $0 < q < \kappa_d$, the negative q-th moments of $M_\xi/\xi^{2/3}$ are uniformly bounded for sufficiently small ξ. Moreover, there are dimensional constants $c_d, C_d > 0$ such that, for every fixed $A \ge 1$,

$$\frac{1}{4}e^{-C_d A^3} \le \liminf_{\xi\downarrow 0} \mathbb{P}[M_\xi > A\xi^{2/3}] \le \limsup_{\xi\downarrow 0} \mathbb{P}[M_\xi > A\xi^{2/3}] \le C_d e^{-c_d A^3}\,. \tag{1.9}$$

Every weak subsequential limit Y of $M_\xi/\xi^{2/3}$ satisfies

$$\begin{aligned} \mathbb{P}[Y \le a] &\le C_d a^{d-1}, \qquad 0 < a \le 1, \\ \frac{1}{4}e^{-C_d A^3} \le \mathbb{P}[Y > A] &\le C_d e^{-c_d A^3}, \quad A \ge 1\,. \end{aligned} \tag{1.10}$$

For a family with the usual γ-parameterization, (1.5) gives precisely the same conclusions with $M_\gamma/\gamma^{2/3}$. In particular, the exponent is $s = 2/3$ in every fixed dimension.

The meaning of nontriviality in (1.8) is exactly tightness and separation from zero in probability. Every subsequential limit is concentrated on $(0, \infty)$ and has unbounded support, so it cannot be deterministic. The theorem does not identify a unique limiting distribution. It also singles out the exponent: if $s < 2/3$, then $M_\gamma/\gamma^s \to 0$ in probability, while for $s > 2/3$ this ratio tends to infinity in probability. For $d = 2$, the lower-tail estimate is linear in a; in particular it implies the $a^{3/5}$ estimate in the planar manuscript [P].

Theorem B. *Properties* (i)–(iv) *imply*

$$-C_d\xi^2 \le 1 - a_\xi \le C_d\xi^{4/3}\,, \tag{1.11}$$

for sufficiently small ξ. They also imply (1.6)*. The proof of these assertions does not use conformal covariance.*

Theorem C. *Parametrize the geodesic $P_\xi : [0,1] \to \mathbb{R}^d$ at constant metric speed and put*

$$X_\xi(t) = \frac{P_\xi(t) - te_1}{\xi^{2/3}}\,. \tag{1.12}$$

For every sequence $\xi_n \to 0$, the laws of X_{ξ_n} are tight in $C([0,1], \mathbb{R}^d)$ with its uniform topology. Moreover,

$$\sup_{t\in[0,1]} |X_{\xi,1}(t)| \longrightarrow 0 \quad \textit{in probability}\,. \tag{1.13}$$

Every subsequential path limit X has zero endpoints, has first coordinate identically zero, and satisfies

$$\mathbb{P}[\|X\|_\infty \le a] \le C_d a^{d-1}, \qquad 0 < a \le 1\,. \tag{1.14}$$

Its norm also satisfies the upper-tail bounds in (1.10)*. In particular it is almost surely not the zero path and its law is not deterministic. The tightness and longitudinal assertions do not use conformal covariance.*

1.3 Proof outline

For the two-segment path through $e_1/2 + Re_2$, the Euclidean excess length is $\sqrt{1+4R^2} - 1 \sim 2R^2$. The difference between its Gaussian arclength integral and that of the straight segment has variance $\pi R + o(R)$; see (5.3). Balancing the cost R^2 with the fluctuation $\xi\sqrt{R}$ suggests $R = \xi^{2/3}$. This balance is not itself a proof: the minimizing path is chosen after observing the field, and the metric has a non-Gaussian remainder that must be controlled at the same scale.

Uniform metric control. Sections 2–3 use the Gaussian entropy argument of [F, Sections 2–3, pp. 4–9]. We retain its quantitative form because later estimates use the geodesic length measure, the explicit Gaussian derivative, and the dependence on the moment order. The only imported moment assumption is fixed-parameter finiteness. The entropy estimate turns that finiteness into bounds uniform as $\xi \downarrow 0$.

Height and distance exponent. The Gaussian derivative of the log distance is the geodesic's normalized length measure. Section 4 bounds the logarithmic energy of its difference from uniform measure on the segment in terms of the height itself. This gives a deterministic-endpoint expansion whose remainder contains the height moment that we seek to bound.

To apply that expansion at a random point of the geodesic, Section 6 approaches each possible point from opposite longitudinal directions for its two endpoint distances. With longitudinal mesh $\tau\ell_n$ and transverse mesh $R\ell_n^{3/4}$, where $\ell_n = \kappa 2^{-n}$, the Euclidean loss is summable and the number of possible edges grows only exponentially in n, independently of R. A sufficiently high fixed moment controls their errors. Near an endpoint, the detour R^2/τ pays for the sum over longitudinal scales.

The first comparison bounds the height by $C(\xi^{2/3} + \sqrt{|1-a_\xi|})$. A midpoint comparison, retaining the powers r^{a_ξ}, bounds $1-a_\xi$ by an arbitrarily small multiple of a squared height moment and its Gaussian fluctuation. Section 7 closes these inequalities together. Tracking the leading $\sqrt{p}$ cost of Gaussian gradients and maxima gives $\limsup_{\xi\downarrow 0} \xi^{-2/3}\|M_\xi\|_p \le C_d p^{1/3}$ and hence the cubic exponential upper tail, at each fixed tail level.

The lower tail. Section 9 applies the imported conformal identity (1.3) to endpoint-fixing Möbius maps. A bend of transverse size R, cut off in a tube of radius R, has squared Cameron–Martin cost $O(Q_\xi^2 R^3)$. The power R^3 is independent of d because the Cameron–Martin space is $\dot{H}^{d/2}$. At $R \asymp Q_\xi^{-2/3}$ the likelihoods have bounded expected supremum. Packing disjoint midpoint cross-sections then gives the small-height exponent $d-1$. A single bend, without fixing $Q_\xi^2 R^3$, gives the matching cubic exponential lower bound for the upper tail.

The entire path. Section 8 uses chains aligned with moving chords to control triangles with three variable endpoints. Dyadic first hits of longitudinal planes provide a transverse modulus of continuity. The finest mesh is larger than the additive metric error and smaller than $\xi^{2/3}$; the interpolation therefore also controls the original constant-speed path, including its possible backtracking.

Notation. Write $B_R = B_R(0)$ for a Euclidean ball and $\|\cdot\|_p$ for the probabilistic L^p norm. All constants may depend on the fixed dimension. A constant with a moment-order subscript may

also depend on that order. Each moment order is fixed before ξ tends to zero. The sufficiently-small-parameter thresholds may also depend on the chosen metric family, through the convergence $a_\xi \to 1$ and the moment-finiteness thresholds in (iv). For $x \neq y$, let

$$\mu_{xy} = \int_0^1 \delta_{(1-t)x+ty}\,\mathrm{d}t, \qquad H_{xy} = \langle h, \mu_{xy}\rangle, \qquad I_{xy} = |x-y|H_{xy}\,.$$

Thus I_{xy} is the Gaussian pairing with Euclidean arclength on the segment; it does not change sign on reversing the endpoints. Set $I_{xx} = 0$, and put $\mu_0 = \mu_{0e_1}$.

2 Gaussian calculus

We first fix the Gaussian norms used to compare line integrals and geodesic length measures. The covariance and differentiation conventions are those of [F, Section 2, pp. 4–5].

Set

$$j_d(x) = \int_{S^{d-1}} \log|x-u|\,\sigma_d(\,\mathrm{d}u), \qquad c_d = j_d(e_1)\,. \tag{2.1}$$

Subtracting the spherical average in both arguments of the logarithmic covariance gives the pinned kernel

$$G_d(x,y) = -\log|x-y| + j_d(x) + j_d(y) - c_d\,. \tag{2.2}$$

Here $c_d = \iint \log|u-v|\,\sigma_d(\,\mathrm{d}u)\sigma_d(\,\mathrm{d}v)$ by rotation invariance. For $d=2$ the mean-value property gives $j_2(x) = \log\max\{1,|x|\}$ and $c_2 = 0$, recovering the usual pinned GFF covariance. The function j_d is continuous, bounded on compact sets, and bounded above by $\log(1+|x|)$. Integrability at the sphere follows, for example, by bounding the sphere mass of a ball of radius r by $C_d r^{d-1}$ and integrating the logarithmic singularity.

For a fixed compact set K, direct integration of (2.2) gives

$$\operatorname{Var} h_r(z) \le \log(1/r) + C_K, \qquad z \in K, \quad 0 < r \le 1, \tag{2.3}$$

$$\operatorname{Var} h_R(0) \le \log R + C_d, \qquad R \ge 1\,. \tag{2.4}$$

For clarity, the exact expression for the first variance is

$$-\log r + 2\int j_d(z+ru)\,\sigma_d(\,\mathrm{d}u) - 2c_d\,.$$

For the second use $j_d(Re_1) = \log R + j_d(R^{-1}e_1)$. Spherical averages and the transformed field in (1.1) need not be independent.

Let $\mathcal{H}_d$ be the Cameron–Martin space of the pinned field. We use a countable orthonormal basis (e_j) with smooth bounded representatives. Such a basis is obtained from a dense family of compactly supported smooth functions modulo constants, by subtracting their sphere averages and applying Gram–Schmidt. For a compactly supported finite signed measure μ whose total variation has finite logarithmic energy,

$$\|\mu\|^2_{\mathcal{H}_d^*} = \iint G_d(x,y)\,\mu(\,\mathrm{d}x)\mu(\,\mathrm{d}y) = \sum_j \left(\int e_j\,\mathrm{d}\mu\right)^2. \tag{2.5}$$

This is the functional on pinned representatives, or equivalently the zero-mass functional $\mu - \mu(\mathbb{R}^d)\sigma_d$. If $\mu(\mathbb{R}^d) = 0$, the one-variable terms in (2.2) cancel. For random measures this identity is applied to each realization, not as a conditional covariance formula.

We shall use the Gaussian inequalities

$$\|F - \mathbb{E}F\|_p \le C\sqrt{p}\,\left\|\|\nabla F\|_{\mathcal{H}_d}\right\|_p, \qquad p \ge 2, \tag{2.6}$$

$$\mathrm{Ent}(e^{tF}) \le \frac{t^2}{2}\mathbb{E}[e^{tF}\,\|\nabla F\|^2_{\mathcal{H}_d}]\,. \tag{2.7}$$

For (2.6) in finite dimensions, integrate $F(X\cos s + Y\sin s)$ over $0 \le s \le \pi/2$, with X, Y independent standard Gaussian vectors. Its tangent vector is independent of its argument. Minkowski and the L^p norm of a standard Gaussian yield the bound. The Gaussian logarithmic Sobolev inequality (2.7) follows from the Ornstein–Uhlenbeck semigroup: for positive smooth g,

$$\mathrm{Ent}(g) = \int_0^\infty \mathbb{E}\frac{|\nabla P_s g|^2}{P_s g}\,\mathrm{d}s \le \int_0^\infty e^{-2s}\mathbb{E}\frac{|\nabla g|^2}{g}\,\mathrm{d}s = \frac{1}{2}\mathbb{E}\frac{|\nabla g|^2}{g}\,.$$

Apply this to $g = e^{tF}$. Conditional expectation onto finitely many Gaussian coordinates, followed by truncation and closure of weak derivatives, gives the displayed inequalities in the present setting. We verify the required integrability before using (2.7).

3 Uniform metric estimates

The fixed-parameter moment theorem supplies finiteness, not bounds uniform as $\xi \downarrow 0$. We use the entropy argument of [F, Section 3, pp. 5–9] to obtain that uniformity. The proof is retained to expose the specific geodesic-energy and moment-order bounds used below. Normalize by the mean logarithmic distance:

$$b_\xi = \mathbb{E}\log D_h^\xi(0, e_1), \qquad \rho_h^\xi = e^{-b_\xi} D_h^\xi, \qquad F_\xi = \log \rho_h^\xi(0, e_1)\,. \tag{3.1}$$

Thus $\mathbb{E}F_\xi = 0$. The number b_ξ is finite by (1.2); no bound on it is used in the bootstrap. Suppress ξ in this section and put

$$K(P) = \mathbb{E}e^{P|F|}, \qquad \psi(t) = \log \mathbb{E}e^{tF}\,.$$

We restrict to parameters for which $3/4 \le a_\xi \le 5/4$.

3.1 Control by an endpoint moment

By (1.1), for $r = |y - x|$,

$$\rho_h(x, y) = r^a e^{\xi h_r(x)} e^{F(h^{x,r,O})}, \qquad Oe_1 = (y - x)/r\,. \tag{3.2}$$

Cauchy–Schwarz and (2.3) give, on each fixed compact set and for $0 < r \le 1$,

$$\mathbb{E}\rho_h(x, y)^p \le C_{p,K} r^{ap - p^2\xi^2} K(2p)^{1/2}\,. \tag{3.3}$$

For r bounded above and away from zero, the corresponding bound for $\mathbb{E}\rho_h(x, y)^{-p}$ is $CK(2p)^{1/2}$. These constants are uniform for sufficiently small ξ at each fixed p.

Let $\alpha = \theta = 1/4$ and define

$$A_R = \sup_{x \ne y \in \overline{B}_R} \frac{\rho(x, y)}{|x - y|^\alpha}, \qquad A = 1 \vee \sup_{n \ge 0} 2^{-2n} A_{2^{n+2}}, \tag{3.4}$$

$$L_n = \rho(\partial B_{2^{n+2}}, \partial B_{2^{n+3}}), \qquad I = 1 \wedge \inf_{n \ge 0} 2^{-\theta n} L_n\,. \tag{3.5}$$

All distances, including set distances, are unrestricted distances in $\mathbb{R}^d$.

Lemma 3.1. *For every fixed $q \geq 1$, there are finite $P = P(q,d)$ and $C = C(q,d)$ such that, for all sufficiently small ξ,*

$$\mathbb{E}[A^q + I^{-q}] \leq C(1 + K(P)) . \tag{3.6}$$

For a fixed compact K and $\delta > 0$, the same type of bound holds for

$$J(K,\delta) = \inf_{x,y \in K:\, |x-y| \geq \delta} \rho(x,y) \quad \text{with } J(K,\delta)^{-q} \text{ in place of } I^{-q} .$$

In particular, all these quantities are finite at each sufficiently small fixed parameter, without yet asserting uniformity in that parameter.

Proof. First let $p \geq 8(d+1)$. On a fixed cube, let M_k be the largest metric distance over neighboring dyadic grid edges at mesh 2^{-k}, using a fixed slightly larger cube. There are at most $C_d 2^{dk}$ such edges. The triangle inequality and approximation of arbitrary endpoints by nested grid points give

$$A_1 \leq C_d \sum_{k \geq 0} 2^{\alpha k} M_k .$$

One obtains this inequality first for grid endpoints and then uses continuity of the metric. Equation (3.3) gives

$$\|M_k\|_p \leq C_p K(2p)^{1/(2p)} 2^{-k(a - p\xi^2 - d/p)} .$$

The exponent $a - \alpha - p\xi^2 - d/p$ is at least $1/4$ when ξ is small enough. Hence

$$\mathbb{E}A_1^p \leq C_p K(2p)^{1/2} . \tag{3.7}$$

Apply similarity covariance at radius $R \geq 1$, then Cauchy–Schwarz and (2.4), to obtain

$$\mathbb{E}A_R^p \leq C_p R^{p(a-\alpha) + p^2\xi^2} K(4p)^{1/4} . \tag{3.8}$$

Since $a - \alpha \leq 1$, Minkowski's inequality and the weights in (3.4) show

$$\mathbb{E}A^p \leq C_p K(4p)^{1/4} . \tag{3.9}$$

For smaller moment orders use monotonicity of L^p norms.

For the lower bound fix K, δ, enlarging K to a fixed ball when convenient. On $\{A \leq t^{-1}\}$, a deterministic net of mesh $c_{K,\delta} t^{2/\alpha} = c_{K,\delta} t^8$ has metric interpolation error at most $t/4$ for each endpoint. Use a slightly larger ball to contain the net. If $J(K,\delta) < t$, some pair of net points has Euclidean separation at least $\delta/2$ and metric distance less than $2t$. The number of possible pairs is at most $C_{K,\delta} t^{-16d}$. Thus, for any fixed $u, v \geq 1$ and all sufficiently small t,

$$\mathbb{P}[J(K,\delta) < t] \leq t^u \mathbb{E}A^u + C_{K,\delta,v} t^{v-16d} K(2v)^{1/2} .$$

Take $u = q+1$ and $v = q + 16d + 1$ and integrate the negative-moment tail. This proves

$$\mathbb{E}J(K,\delta)^{-q} \leq C\left(1 + \mathbb{E}A^{q+1} + K(2q + 32d + 2)^{1/2}\right) . \tag{3.10}$$

No restriction on the paths is used in this net argument.

Let $L = \rho(\partial B_1, \partial B_2)$. Applying (3.10) to a ball containing these spheres bounds every prescribed inverse moment of L in terms of some $K(P)$. By scaling and Cauchy–Schwarz,

$$\left\| L_n^{-1} \right\|_q \leq C_q 2^{-n(a - q\xi^2)} (\mathbb{E}L^{-2q})^{1/(2q)} .$$

Since $a - \theta - q\xi^2 > 0$ for small enough ξ, sum $2^{\theta n} \left\| L_n^{-1} \right\|_q$ over n. This proves the claim for I. All uses of $K(P)$ involve only finitely many fixed moment orders, which are finite by (1.2). □

3.2 Geodesic energy and differentiation

Set

$$U = 1 + \log A + \log I^{-1} + F_-, \qquad F_- = \max\{-F, 0\}\,. \tag{3.11}$$

There is a fixed $P_0 = P_0(d) < \infty$ such that

$$\Lambda := \log \mathbb{E}e^U \le C_d + C_d(\psi(P_0) + \psi(-P_0))\,. \tag{3.12}$$

Here and below Λ is finite before we prove it is uniformly bounded. To verify (3.12), apply Hölder to $e^U = eAI^{-1}e^{F_-}$. It uses only $\mathbb{E}A^3$, $\mathbb{E}I^{-3}$, and $K(3)$. The estimate for I^{-3} uses $\mathbb{E}L^{-6}$; (3.10) then uses $\mathbb{E}A^7$ and $K(32d+14)$. The upper estimates need at most $K(32(d+1))$. Consequently $P_0 = 64(d+1)$ is sufficient. Finally,

$$\log K(P_0) \le \log 2 + \psi(P_0) + \psi(-P_0)\,,$$

since $\mathbb{E}F = 0$ implies $\psi(t) \ge 0$.

We next obtain a bound for ∇F involving this same U. By Lemma 3.1, $I > 0$ almost surely. The annular barriers $L_n \ge I2^{\theta n}$ therefore tend to infinity. Every bounded metric ball is Euclidean bounded; because the metric induces the Euclidean topology, its closed bounded balls are compact. A proper length space is geodesic.

Every geodesic from 0 to e_1 has length $e^F \le A$. If it leaves $B_{2^{n+3}}$, it must cross the annulus defining L_n. Hence all these geodesics lie in B_T, where

$$T \le C_d(A/I)^4\,. \tag{3.13}$$

For any such geodesic P, parametrized at constant metric speed on $[0,1]$, write

$$\nu_P = \int_0^1 \delta_{P(t)}\,\mathrm{d}t\,.$$

Between the first and last visits to a closed Euclidean ball of radius $r \le 1$, the path is still a geodesic. The upper estimate supplied by A therefore gives

$$\nu_P(B_r(z)) \le Br^\alpha, \qquad B = 1 \vee (C_d A(1+T)^2 e^{-F})\,. \tag{3.14}$$

Repeated visits cause no problem, since all visits occur between those first and last times. Equations (3.11)–(3.14) imply

$$\log B + \log(1+T) \le C_d U\,.$$

Moreover,

$$\iint \log^+ \frac{1}{|x-y|}\,\nu_P(\mathrm{d}x)\nu_P(\mathrm{d}y) \le \int_0^\infty \min\{1, Be^{-\alpha s}\}\,\mathrm{d}s \le \alpha^{-1}(1+\log B)\,.$$

By (2.2) and (2.5),

$$\|\nu_P\|^2_{\mathcal{H}_d^*} \le C_d U\,. \tag{3.15}$$

Weyl scaling makes $\log D_h^\xi(0,e_1)$ Lipschitz along each Gaussian coordinate line, with constant $\xi\,\|e_j\|_\infty$. At a point of coordinate differentiability, an unperturbed geodesic is a competitor for the shifted metric, so

$$F(h+te_j) - F(h) \le \log \int e^{\xi t e_j}\,\mathrm{d}\nu_P = \xi t \int e_j\,\mathrm{d}\nu_P + O(t^2)\,.$$

Divide by positive and by negative t. The two bounds identify the derivative. Fubini and a countable intersection therefore give, almost surely for all j,

$$\partial_j F = \xi \int e_j \, \mathrm{d}\nu_P, \qquad \|\nabla F\|_{\mathcal{H}_d}^2 \le C_d \xi^2 U \,. \tag{3.16}$$

The derivative identity holds for every geodesic at such a field. A measurable choice is available: constant-speed geodesics have nonempty compact fibers in path space, with a Borel graph. Alternatively, the bound in (3.16) follows using any geodesic in each realization, since it is uniform over the choice. The fixed-parameter moments give $F \in L^2$, and the sum of the squared coordinate derivatives is integrable by (3.16). To see the weak derivative explicitly, set $g_j = \partial_j F$, let $\mathcal{F}_n$ be the sigma field of the first n Gaussian coordinates, and put $F_n = \mathbb{E}[F \mid \mathcal{F}_n]$. Coordinate integration by parts gives $\partial_j F_n = \mathbb{E}[g_j \mid \mathcal{F}_n]$ for $j \le n$. The functions F_n converge to F in L^2, and their projected vector-valued gradients converge to $(g_j)_j$ in $L^2(\ell^2)$. Closure of the Gaussian gradient gives the asserted weak derivative.

3.3 Closing the entropy estimate

All numbers in (3.12) are finite at each sufficiently small fixed ξ. Repeating only the finiteness part of Lemma 3.1 at finitely many larger fixed orders, we may also assume $\mathbb{E}e^{(2P_0+2)|F|} + \mathbb{E}e^{2U} < \infty$. Hölder then gives $\mathbb{E}[e^{tF}U] < \infty$ for $|t| \le P_0$. The coordinate derivatives in (3.16) are weak Gaussian derivatives. Applying the chain rule to truncated exponentials and using this integrability puts $e^{tF/2}$ in the Gaussian Sobolev space. Thus (2.7) applies. These extra moments justify the calculation; their values will not occur in its bounds.

For $0 \le t \le P_0$, let $\mathbb{E}_t$ denote expectation tilted by $e^{tF-\psi(t)}$, and set

$$\mathcal{E}(t) = t\psi'(t) - \psi(t) \,.$$

The logarithmic Sobolev inequality and (3.16) imply

$$\mathcal{E}(t) \le \frac{C_d \xi^2 t^2}{2} \mathbb{E}_t U \,. \tag{3.17}$$

The entropy variational inequality, applied to this probability tilt, gives

$$\mathbb{E}_t U \le \mathcal{E}(t) + \log \mathbb{E}e^U = \mathcal{E}(t) + \Lambda \,. \tag{3.18}$$

For example, (3.18) follows by Jensen applied under the tilted law to $e^U/e^{tF-\psi(t)}$. If ξ is small enough that $C_d\xi^2 P_0^2/2 \le 1/2$, absorb the entropy term to obtain

$$\mathcal{E}(t) \le C_d \xi^2 t^2 \Lambda \,.$$

Since $\psi(0) = \psi'(0) = 0$, integration of $(\psi(t)/t)' = \mathcal{E}(t)/t^2$ yields

$$\psi(t) \le C_d \xi^2 t^2 \Lambda, \qquad 0 \le t \le P_0 \,.$$

The identical argument for $-F$ gives the bound at negative t. Substitution in (3.12) now yields

$$\Lambda \le C_d + C_d \xi^2 P_0^2 \Lambda \,.$$

Decrease the upper bound on ξ once more. Since $\Lambda < \infty$ was already established, this is a valid absorption between finite numbers, and proves

$$\sup_{0<\xi\le\xi_0} \mathbb{E}e^{U_\xi} < \infty \,. \tag{3.19}$$

This is the step which upgrades fixed-parameter finiteness to uniform estimates.

Apply (2.6) first to a bounded 1-Lipschitz truncation $T_N(F)$, centered by its expectation. Its gradient is bounded by that of F. Since $F \in L^1$, these expectations converge to zero; Fatou's lemma then permits $N \to \infty$ without assuming an L^p bound for F in advance. Equations (3.16) and (3.19) give

$$\|F_\xi\|_p \le C_d p \xi, \qquad p \ge 2, \quad 0 < \xi \le \xi_0 \,. \tag{3.20}$$

The constants in this display do not depend on p: $\left\|\sqrt{U}\right\|_p \le C_d\sqrt{p}$ follows from (3.19). Expanding the exponential in a series and using $k! \ge (k/e)^k$ shows that for some $c > 0$,

$$\sup_{0<\xi\le\xi_0} \mathbb{E}\exp(c|F_\xi|/\xi) \le 2 \,. \tag{3.21}$$

In particular, $K(P)$ is uniformly bounded for every fixed P, after reducing the upper bound on ξ if necessary. Returning to the moment estimates proves the following conclusion.

Proposition 3.2. *For each fixed $p < \infty$, $\mathbb{E}[A_\xi^p + I_\xi^{-p}]$ is bounded uniformly for sufficiently small ξ. The same is true for every fixed compact separated-distance inverse moment $\mathbb{E}J_\xi(K,\delta)^{-p}$. More generally, for each fixed $\beta \in (0,1)$,*

$$\mathbb{E}\left[\left(\sup_{x\ne y\in K} \frac{\rho_h^\xi(x,y)}{|x-y|^\beta}\right)^p\right] \le C_{K,\beta,p} \,, \tag{3.22}$$

uniformly for sufficiently small ξ.

Proof. The first assertions follow from Lemma 3.1 and (3.21). For (3.22), repeat the dyadic proof of (3.7) with β in place of α: choose a fixed edge moment order $q \ge p$ so large that $d/q < (1-\beta)/4$, and then ξ so small that $a_\xi - \beta - q\xi^2 - d/q > 0$. Scaling covers a fixed compact set. □

4 The first-order remainder

We continue to use the mean-log normalization $\rho = e^{-\mathbb{E}\log D_h^\xi(0,e_1)} D_h^\xi$ and $F = \log\rho(0,e_1)$. Let $P : [0,1] \to \mathbb{R}^d$ be a geodesic at constant metric speed, and write

$$M = \max_t \operatorname{dist}(P(t),[0,e_1]), \qquad \nu = \int_0^1 \delta_{P(t)}\,\mathrm{d}t, \qquad m_p = \|M\|_{L^p} \,.$$

Existence was proved in Section 3. The bound (3.13) and Proposition 3.2 give $\sup_{\xi\le\xi_p}\mathbb{E}M^p < \infty$ for every fixed p.

We will need uniqueness of a minimizing path between fixed endpoints. This is distinct from uniqueness of the metric assignment imported in Section 1; it follows here from directional differentiation of the distance.

Lemma 4.1. *The geodesic between any fixed pair of distinct deterministic points is almost surely unique. In any fixed connected open domain, there is almost surely at most one internal geodesic between such a pair, whenever an internal minimizer exists.*

Proof. Choose a countable family of smooth compactly supported functions separating finite Borel measures, and replace each by its sphere-pinned representative f. These representatives belong to $\mathcal{H}_d$ and still separate probability measures. For each such f, Weyl scaling makes $t \mapsto \log D_{h+tf}^\xi(x,y)$

Lipschitz. It is almost surely differentiable at $t = 0$: Fubini gives this at almost every t, and Cameron–Martin equivalence transfers the conclusion to zero. Testing any original geodesic as a competitor, for both signs of t, identifies the derivative with $\xi \int f \, \mathrm{d}\nu_P$, exactly as in (3.16). Thus all geodesics have the same normalized length measure. Its support is the whole geodesic image. A geodesic with distinct endpoints is a simple arc, and the image with its ordered endpoints determines its constant-speed parametrization. For the internal assertion, use $\log D_h^\xi(x, y; U)$ in the same argument. This distance is finite and positive: locally, sufficiently short almost minimizing ambient paths remain in U, and a finite chain of such neighborhoods connects any two points of U. The directional Lipschitz bound holds even when no minimizer exists; if one exists, the competitor argument applies to every minimizer. □

Put

$$\lambda = 1 - a_\xi, \quad \ell = |\lambda|, \quad L = \log(e/\xi), \quad \omega = \xi L^2 + \ell \log(e/\ell), \quad \delta = \xi^{2/3}, \tag{4.1}$$

with the term containing ℓ defined to be zero at $\ell = 0$. Only $\ell \to 0$ is used until the final absorption.

Lemma 4.2. *For each fixed $K < \infty$, there is $C_K < \infty$ such that*

$$\mathbb{P}(G_K^c) \le C_K \xi^K, \qquad G_K = \{\Delta \le C_K(\xi L + \ell)\}, \quad \Delta = \sup_{x,y \in \overline{B}_3} |\rho(x, y) - |x - y|| . \tag{4.2}$$

On G_K, for sufficiently small ξ, every 0-to-e_1 geodesic lies in B_2, and

$$|z| + |z - e_1| - 1 \le 3\Delta, \qquad z \in P . \tag{4.3}$$

Consequently $m_p \to 0$ for every fixed p.

Proof. Take a deterministic grid of mesh ξ^4 in a fixed cube containing B_3. There are at most $C_d \xi^{-8d}$ endpoint pairs. For each distinct grid pair at distance r, (3.2) gives

$$\log \rho(x, y) = a_\xi \log r + \xi h_r(x) + F(h^{x,r,O}) .$$

The Gaussian variable $h_r(x)$ has variance at most $C+4L$. The exponential bound (3.21) and a union bound therefore show, with complement probability at most $C_K \xi^K$, that $|\xi h_r(x) + F(h^{x,r,O})| \le C_K \xi L$ for every grid pair. For a_ξ in a fixed neighborhood of one,

$$\sup_{0<r\le C} |r^{a_\xi} - r| \le C|\lambda| .$$

This estimate follows by differentiating r^a in a; the product $r^a |\log r|$ is bounded uniformly on this range. It has no extra factor L. Hence the additive error on the grid is $C_K(\xi L + \ell)$.

Use the compact Hölder estimate (3.22) with exponent $1/2$. Its constant is at most ξ^{-1} outside an event of probability $C_K \xi^K$, by a sufficiently high fixed moment. Moving each endpoint to the grid costs at most $C\xi$. This proves (4.2). All moment orders used here are fixed after K.

If a geodesic first reached ∂B_2, its distance from zero would be at least $2 - \Delta$, whereas its full length is at most $1 + \Delta$. For $\Delta < 1/2$ this is impossible. The triangle equality along the geodesic then proves (4.3). The corresponding Euclidean ellipse has distance at most $C\sqrt{\Delta}$ from the segment when Δ is small. Combining this bound on G_K with the previously proved uniform $2p$-th moment of M on G_K^c gives $m_p \to 0$. □

Lemma 4.3 (Energy of the geodesic measure)**.** *For each fixed $q \geq 2$,*

$$\left\| \|\nu - \mu_0\|_{\mathcal{H}_d^*}^2 \right\|_{L^{q/2}} \leq C_q(m_{q/2} + \omega)\,. \tag{4.4}$$

In particular, if

$$Z = F - \xi\langle h, \mu_0\rangle\,,$$

then $\mathbb{E}Z = 0$ and

$$\|Z\|_{L^q} \leq C_q \xi (m_{q/2} + \omega)^{1/2}\,. \tag{4.5}$$

Proof. We give the deterministic logarithmic-energy comparison. On $\{\Delta < 1/10\}$, the constant-speed identities imply

$$\big||P(t)| - t\big| \leq C\Delta, \qquad |P(t) - te_1| \leq C(M + \Delta)\,.$$

For the second inequality, project onto the segment and use the definition of M; it also covers a projection beyond an endpoint. Couple ν and μ_0 by the common parameter t and put $u = C(M+\Delta)$. If $u = 0$, the coupled measures are equal and the energy is zero. Hence assume $u > 0$ below, with the convention $\Delta|\log\Delta| = 0$ when $\Delta = 0$. Between the first and last visits to any Euclidean ball of radius r, geodesicity and the additive metric bound give

$$\nu(B_r(x)) \leq C(r + \Delta)\,. \tag{4.6}$$

The Hölder estimate also gives

$$\nu(B_r(x)) \leq \mathcal{A} r^{1/2}\,, \tag{4.7}$$

where $\mathcal{A} \geq 1$ has uniformly bounded fixed positive moments. We may take a constant times the compact Hölder constant plus one, since $\rho(0, e_1) \geq 1/2$ on this event.

The one-variable terms of the pinned kernel cancel for $\nu - \mu_0$. Write its energy as the four-kernel integral over two copies of the coupling, calling the geodesic-side points $x = P(t)$ and $y = P(s)$. When $|x - y| > 4u$, the mixed Hessian bound for $-\log|x - y|$ gives an integrand bounded in absolute value by $Cu^2/|x - y|^2$. Summing Euclidean annuli with (4.6) bounds this part by Cu. In the remaining region all four distances are at most $6u$. Subtract $\log(1/u)$ from the four logarithmic kernels; these constants cancel. Their negative parts are bounded and have mass $O(u + \Delta)$. The two positive mixed terms occur with negative signs and can be discarded in an upper bound. For the self terms, layer cake and (4.6)–(4.7) give

$$\int_0^{Cu} \min\{1, C(r + \Delta), \mathcal{A} r^{1/2}\}\,\frac{\mathrm{d}r}{r} \leq Cu + C\Delta(1 + \log\mathcal{A} + |\log\Delta|)\,.$$

The same estimate for μ_0 follows from its linear ball growth. Truncation justifies the four-kernel computation at the diagonal. Thus

$$\|\nu - \mu_0\|_{\mathcal{H}_d^*}^2 \leq CM + C\Delta(1 + \log\mathcal{A} + |\log\Delta|)\,. \tag{4.8}$$

The same layer-cake calculation covers u bounded away from zero, since the integral over $r \geq \Delta$ is bounded by $C(u + \Delta)$.

Apply (4.8) on G_K. For $b = \xi L + \ell$, the elementary bound $b\log(e/b) \leq C\omega$ and the bounded moments of $\log\mathcal{A}$ give an $L^{q/2}$ error at most $C_q\omega$. On G_K^c, (3.15) bounds the energy by $C(U + 1)$. Hölder and (3.19) make its $L^{q/2}$ contribution at most $C_q\xi^{K/q}$; choose K large enough to absorb it into ω. This proves (4.4).

The derivative formula (3.16) identifies the Gaussian gradient of Z with $\xi(\nu - \mu_0)$ in the Cameron–Martin dual. Both terms defining Z are centered. Apply (2.6) and (4.4) to obtain (4.5). □

Lemma 4.4 (Expansion at deterministic endpoints)**.** *Fix $q \geq 2$, a compact endpoint set and $0 < \alpha < 1$. For all sufficiently small ξ and all distinct endpoints in that set, with $r = |x - y|$,*

$$\|\rho(x,y) - r - \xi I_{xy}\|_{L^q} \leq C_{q,\alpha} r^{\alpha} (\xi \sqrt{m_{q/2} + \omega} + \ell + \xi^2), \tag{4.9}$$

$$\|\rho(x,y) - r^{a_\xi} - \xi I_{xy}\|_{L^q} \leq C_{q,\alpha} r^{\alpha} (\xi \sqrt{m_{q/2} + \omega} + \xi\ell + \xi^2)\,. \tag{4.10}$$

The terms involving $m_{q/2}$ have exactly this moment order.

Proof. By the exact similarity rule, the variable

$$Z_{xy} = \log \rho(x,y) - a_\xi \log r - \xi H_{xy}\,,$$

has the law of Z. Indeed, $H_{xy} = h_r(x) + H_{0e_1}(h^{x,r,O})$, so the spherical-average term cancels. No independence is asserted. Set $X = \xi H_{xy} + Z_{xy} = \xi h_r(x) + F(h^{x,r,O})$. Exactly,

$$\rho(x,y) = r^{a_\xi} \left(1 + \xi H_{xy} + Z_{xy} + \left(e^X - 1 - X\right)\right)\,. \tag{4.11}$$

Keep $r^{a_\xi} Z_{xy}$ as a linear term. The basic estimates for F and the Gaussian spherical average imply

$$\left\| e^X - 1 - X \right\|_{L^q} \leq \tfrac{1}{2} \left\| X^2 e^{|X|} \right\|_{L^q} \leq C_q \xi^2 (1 + |\log r|)\, r^{-C_q \xi^2}, \qquad 0 < r \leq 1\,.$$

To verify the last inequality, use $\|X\|_{4q} \leq C_q \xi (1 + \sqrt{|\log r|})$ and $\left\| e^{|X|} \right\|_{2q} \leq C_q r^{-C_q \xi^2}$, with Cauchy–Schwarz for the two possibly dependent summands of X. The exponential bound for F is used only at fixed orders here. For r bounded away from zero the same estimate holds with constants.

For a_ξ sufficiently close to one, powers of $|\log r|$ are absorbed by an arbitrarily small fixed loss in the power of r. The deterministic difference $r^{a_\xi} - r$ costs $C\ell r^{\alpha}$, and the change from $\xi r^{a_\xi} H_{xy}$ to $\xi r H_{xy}$ costs $C_q \xi \ell r^{\alpha}$. Applying (4.5) to the linear term in (4.11) proves both bounds. In particular, no Hölder inequality is applied to the M-dependent remainder, so its moment order is not increased. □

5 Gaussian line integrals

The random advantage of a bent path is measured by the difference of its line integral and that of its chord. We estimate this difference through logarithmic energy. Only the covariance of the field and Euclidean geometry enter this section; conformal covariance of the metric is not used.

Lemma 5.1 (Transport in logarithmic energy)**.** *Let μ and ν be positive measures of the same mass at most C, supported in a fixed compact set. Suppose $\mu(B_r(x)) + \nu(B_r(x)) \leq Cr$ and that they have a coupling supported on $|x - y| \leq u$, where $0 < u \leq 1$. Then*

$$\|\mu - \nu\|^2_{\mathcal{H}^*_d} \leq C' u\,. \tag{5.1}$$

Proof. The one-variable pinning terms cancel. Write the energy as the four-kernel integral over two copies of the coupling. Where the original points are separated by more than $4u$, the mixed Hessian bound for the logarithmic kernel gives $Cu^2/|x - x'|^2$. The linear ball bound makes the sum over annuli $O(u)$. On the complementary region all four distances are at most $6u$. Subtract

$\log(1/u)$ in every kernel. The negative parts are bounded; the positive mixed terms have negative signs and can be discarded. Each remaining self term is bounded by

$$\int_0^\infty \mu \otimes \mu\{|x - x'| < ue^{-s}\}\, \mathrm{d}s \le C \int_0^\infty ue^{-s}\, \mathrm{d}s = Cu\,,$$

and likewise for ν. First truncate the diagonal singularities and then pass to the limit, using these same integrable bounds. □

Lemma 5.2 (A triangle of line integrals)**.** *Let a, b, c lie in a fixed compact set. Write $s = |a-c| > 0$ and $H = \operatorname{dist}(b, [a, c])$. Then*

$$\|I_{ab} + I_{bc} - I_{ac}\|_{L^2} \le C\sqrt{sH} + CH\sqrt{\log(e + 1/s)}\,. \tag{5.2}$$

The line integrals I_{xy} have a version continuous jointly in their endpoints, including $I_{xx} = 0$.

Proof. Let P be the broken line $a \to b \to c$ and let $S = [a, c]$. Its length s_P satisfies $s_P - s \le 2H$, by comparing through the point of S nearest to b. At the normalized arclength time $t_b = |a-b|/s_P$, the distance from b to $a + t_b(c - a)$ is at most $4H$. For example, if the nearest point has arclength coordinate u on S, then

$$\left| u - \frac{s|a - b|}{s_P} \right| \le \big| u - |a - b| \big| + |a - b| \frac{s_P - s}{s_P} \le 3H\,.$$

Both constant-speed curves are affine on either side of t_b. Their normalized arclength measures therefore admit a displacement coupling of size $4H$. When $H \le s$, both measures, after rescaling space by s, have a uniform linear ball bound and bounded support. Lemma 5.1 gives energy at most CH/s for their difference when $4H \le s$. For $s/4 < H \le s$, the uniform logarithmic self-energy bound after rescaling gives the same estimate with a larger constant. The arclength measures satisfy

$$\operatorname{arc}(P) - \operatorname{arc}(S) = s_P(\mu_P - \mu_S) + (s_P - s)\mu_S\,.$$

The first term has squared norm at most CsH. Direct integration on S bounds $\|\mu_S\|^2_{\mathcal{H}_d^*} \le C \log(e + 1/s)$, giving the second term in (5.2). If $H > s$, both edges of P have length at most $3H$. Their line-integral norms are at most $CH\sqrt{\log(e + 1/H)}$, by direct integration of the pinned kernel. This is bounded by the second term in (5.2).

For continuity, put $v = |x - x'| + |y - y'| \le 1/4$ and $r = |x - y|$. If one segment has length at most $2\sqrt{v}$, both line-integral norms are at most $C\sqrt{v}\sqrt{\log(e/v)} \le Cv^{1/4}$. Otherwise the lengths are comparable and at least $\sqrt{v}$. Coupling the normalized segment measures at common affine time gives displacement v. The same scaled transport estimate, together with the difference of their masses, bounds the L^2 difference by $C\sqrt{v} + Cv\sqrt{\log(e/v)} \le Cv^{1/4}$. Gaussian moments upgrade this to an L^q bound for every q. Taking $q > 8d$ and applying the usual dyadic-net proof of the continuity criterion constructs a continuous version. We use this version throughout. □

For a midpoint bend, the square-root fluctuation is exact. If $z_R = e_1/2 + Re_2$, then

$$\operatorname{Var}\left(I_{0z_R} + I_{z_R e_1} - I_{0e_1}\right) = \pi R + o(R), \qquad R \downarrow 0\,. \tag{5.3}$$

Indeed, let $f(t) = 2\min\{t, 1 - t\}$ and let μ_R be the pushforward of $\mathrm{d}t$ under $t \mapsto te_1 + Rf(t)e_2$. The signed measure $\mu_R - \mu_0$ has mass zero, so its energy is independent of the pinning terms.

Expanding its four logarithmic kernels and using symmetry gives

$$E_R := \|\mu_R - \mu_0\|^2_{\mathcal{H}_d^*} = \int_0^1\int_0^1 \log\left(1 + \frac{R^2 f(t)^2}{(t-s)^2}\right) \mathrm{d}s\,\mathrm{d}t \\ - \frac{1}{2}\int_0^1\int_0^1 \log\left(1 + R^2\frac{(f(t)-f(s))^2}{(t-s)^2}\right) \mathrm{d}s\,\mathrm{d}t\,.$$

The second integral is $O(R^2)$, since f is Lipschitz. In the first, put $s = t + Ru$ and use $\int_{\mathbb{R}} \log(1 + a^2/u^2)\,\mathrm{d}u = 2\pi|a|$. Dominated convergence gives $E_R/R \to 2\pi\int_0^1 f(t)\,\mathrm{d}t = \pi$. Both edges of the bent path have the same speed in this parameterization, so its arclength measure is $\sqrt{1+4R^2}\,\mu_R$. The measures μ_R have uniformly bounded logarithmic self-energy, by their uniform linear ball growth. Replacing μ_R by this measure changes the norm of its difference from μ_0 by $O(R^2)$. Since $E_R = O(R)$, the squared norm changes by $O(R^2)O(\sqrt{R}) + O(R^4) = o(R)$, proving (5.3).

We will repeatedly use the elementary Gaussian maximum estimate

$$\left\| \max_{1\le j\le N} |X_j| \right\|_{L^q} \le C\sigma\sqrt{q + \log(2N)}, \qquad q \ge 1\,, \tag{5.4}$$

for centered Gaussian variables with variances at most σ^2. It follows by integrating the union bound $\mathbb{P}[\max_j |X_j| > t] \le 2N\exp(-t^2/(2\sigma^2))$. No independence is required.

6 Random endpoints

The metric expansion is available at deterministic pairs. We now construct a countable set of pairs which controls all points in a thin cylinder, while spending only a small fraction of the Euclidean detour.

Fix $\beta = \alpha = 3/4$. The longitudinal intervals are

$$J_0 = [1/4, 3/4], \quad \tau_0 = 1, \qquad J_k = [2^{-k-1}, 2^{-k}], \quad \tau_k = 2^{-k}, \quad k \ge 2\,, \tag{6.1}$$

together with the reflections $1 - J_k$. They cover $(0,1)$. For one such interval J with scale τ, and $0 < R \le \tau$, put

$$\mathcal{C}(J,R) = \{(t,v) : t \in J,\ v \perp e_1,\ |v| \le 2R\}\,.$$

We only use R below a fixed small constant.

Lemma 6.1 (Endpoint chains)**.** *Fix $\eta > 0$ and a fixed $q \ge 2$ with $q > 32d/9$. There are nonnegative random variables $U_{J,R}$ and $V_{J,R}$ such that, simultaneously for $z \in \mathcal{C}(J,R)$,*

$$\rho(0,z) + \rho(z,e_1) - \rho(0,e_1) \ge |z| + |z - e_1| - 1 - \eta R^2/\tau - U_{J,R} - \xi V_{J,R}\,, \tag{6.2}$$

and

$$\|U_{J,R}\|_{L^q} \le C_{q,\eta}(\xi\sqrt{m_{q/2} + \omega} + \ell + \xi^2), \tag{6.3}$$

$$\|V_{J,R}\|_{L^q} \le C_{q,\eta}\sqrt{R}\,. \tag{6.4}$$

The constants are independent of R, τ and the choice of interval.

Proof. Step 1: The deterministic chains. Choose a small number $\kappa > 0$, to be fixed in terms of η and d, and set $\ell_n = \kappa 2^{-n}$ and $h_n = \tau \ell_n$. For $z = (t, v)$, let v_n be the componentwise nearest lattice point to v at mesh $R\ell_n^\beta$. Define

$$t_n^+ = h_n(\lfloor t/h_n \rfloor + 3), \qquad t_n^- = h_n(\lceil t/h_n \rceil - 3), \qquad z_n^\pm = (t_n^\pm, v_n)\,.$$

Both sequences converge to z. The plus sequence decreases longitudinally and the minus sequence increases; their successive longitudinal increments lie in $[h_n, 3h_n/2]$. Also,

$$|v_n - v| \le C_d R\ell_n^\beta, \quad |v_n - v_{n+1}| \le C_d R\ell_n^\beta, \quad |z_n^\pm - z_{n+1}^\pm| \le C_d \tau \ell_n^\beta\,. \tag{6.5}$$

For κ small, all points stay in a fixed compact set, their longitudinal distances to each endpoint are at least $c\tau$, and the endpoint nearer J has chord length comparable to τ. There are at most $C_{\kappa,d}2^{dn}$ possible nodes at level n and at most $C_{\kappa,d}2^{2dn}$ possible edges. These counts do not depend on R or τ.

For the distance from zero use the plus chain; for the distance from e_1 use the minus chain. We check the first. Write $w = v_n - v_{n+1}$ and $h = t_n^+ - t_{n+1}^+ > 0$. Convexity of the Euclidean norm gives

$$|z_{n+1}^+| + |z_n^+ - z_{n+1}^+| - |z_n^+| \le C\left(\frac{|w|^2}{h} + \frac{R^2}{\tau^2}h\right) \le C_d \frac{R^2}{\tau}\ell_n^{2\beta-1}\,.$$

For the first inequality, subtract the supporting linear functional at z_{n+1}^+, use $\sqrt{h^2 + |w|^2} - h \le |w|^2/(2h)$ and absorb the mixed term by $|w|^2/h + (R/\tau)^2 h$. This argument permits arbitrary slopes on the short edge. Since $2\beta - 1 > 0$, summation and telescoping give

$$|z_0^+| - \sum_{n\ge 0} |z_n^+ - z_{n+1}^+| \ge |z| - C_d \frac{R^2}{\tau}\kappa^{2\beta-1}\,. \tag{6.6}$$

Reflection about $t = 1/2$ gives the corresponding estimate for the minus chain anchored at e_1. Choose κ so that their combined cost is at most $\eta R^2/\tau$.

Step 2: The metric errors. Let $E_{xy} = \rho(x, y) - |x - y| - \xi I_{xy}$. The maximum absolute coarse-edge error has L^q norm at most a constant times $B_q = \xi\sqrt{m_{q/2} + \omega} + \ell + \xi^2$, by Lemma 4.4 and the finite coarse count. At level n, the maximum edge error has norm at most

$$C_{q,\kappa} B_q\, 2^{2dn/q}(\tau\ell_n^\beta)^\alpha\,.$$

Choose q so that $2d/q < \alpha\beta$. The sum over n is at most $C_{q,\kappa}B_q$. The same calculation with the basic point-distance moment bound shows that the sum of the maximal edge distances is finite almost surely. The reverse triangle inequality, first for finite chains and then by metric continuity, is therefore legitimate:

$$\rho(0, z) \ge \rho(0, z_0^+) - \sum_{n\ge 0} \rho(z_n^+, z_{n+1}^+)\,.$$

It is essential that the two anchored distances use opposite longitudinal chains.

Step 3: The Gaussian errors. The triangle with vertices $0, z_{n+1}^+, z_n^+$ has distance of its middle vertex from the long chord at most $C_d R\ell_n^\beta$. Indeed, compare z_{n+1}^+ with $(t_{n+1}^+/t_n^+)z_n^+$ and use (6.5). The corresponding statement holds for the triangle anchored at e_1. Each long chord has length comparable either to τ or to one. Lemma 5.2, $R \le \tau$, and $\sup_{0<\tau\le 1} \tau \log(e/\tau) < \infty$ imply that every such triangle defect has variance at most $C_d R\ell_n^\beta$. For example, the second squared term

in (5.2) is at most $CR^2\ell_n^{2\beta}\log(e/\tau) \le CR\ell_n^{\beta}$. The Gaussian maximum estimate and the edge count give

$$\Big\| \sum_{n\ge 0} \max_{\text{level } n} |\text{triangle defect}| \Big\|_{L^q} \le C_{q,\kappa}\sqrt{R}\sum_{n\ge 0} 2^{-\beta n/2}\sqrt{n+q} \le C_{q,\kappa}\sqrt{R}\,.$$

Telescoping the defects reconstructs I_{0z} or I_{ze_1} in the reverse-triangle expansions.

It remains to bound the full triangle field $T(z) = I_{0z} + I_{ze_1} - I_{0e_1}$. Use a plus chain for this Gaussian calculation alone. The coarse variables $T(z_0^+)$ have variance at most CR, since the coarse point has distance at most CR from $[0, e_1]$. For $a = z_n^+$ and $b = z_{n+1}^+$, the increment is

$$T(b) - T(a) = (I_{0b} + I_{ba} - I_{0a}) - (I_{1a} + I_{ab} - I_{1b})\,.$$

Here I_{1a} means $I_{e_1 a}$. The two triangles have opposite orientations: their middle vertices are b and a, respectively. Both altitudes have the preceding bound, and the short-edge line integrals cancel exactly. The same maximum estimate therefore gives

$$\Big\| \sup_{z\in\mathcal{C}(J,R)} |T(z)| \Big\|_{L^q} \le C_{q,\kappa}\sqrt{R}\,.$$

All assertions are first obtained on a fixed dense countable set. The continuous version of the line-integral process and the continuity of ρ extend them to every point of the cylinder, including a field-dependent point. Adding the two metric inequalities, subtracting the expansion of $\rho(0, e_1)$, and collecting absolute errors proves (6.2)–(6.4). □

7 Height and exponent bounds

We first allow an error involving $|1 - a_\xi|$ in the height estimate. The midpoint comparison then controls that error using the height itself. The constants in the first comparison are fixed before the second comparison is made arbitrarily accurate.

Lemma 7.1. *For every $\varepsilon > 0$ there is $C_\varepsilon < \infty$ such that, for $0 < \xi < 1/2$ and $0 \le u \le 1/2$,*

$$\xi\sqrt{u\log(e/u)} \le \varepsilon u + C_\varepsilon \xi^2 \log(e/\xi)\,. \tag{7.1}$$

In particular, with $S = \delta + \sqrt{\ell}$,

$$\xi\sqrt{m_p + \omega} + \ell + \xi^2 \le C\big(\xi\sqrt{m_p} + S^2\big)\,, \tag{7.2}$$

for sufficiently small ξ.

Proof. If $u \le \xi^2$, the left side of (7.1) is at most $C\xi^2\sqrt{\log(e/\xi)}$. If $u > \xi^2$, then $\log(e/u) \le 2\log(e/\xi)$ and the usual quadratic Young inequality gives (7.1). For the last assertion also use $\xi^{3/2}L = o(\xi^{4/3})$ and $\xi^2 L = o(\xi^{4/3})$. □

Proposition 7.2 (A first height bound)**.** *For every fixed $p \ge 4(d+1)$, there is $C_p < \infty$ such that*

$$m_p \le C_p(\delta + \sqrt{\ell})\,, \tag{7.3}$$

for sufficiently small ξ.

Proof. Set $q = 2p$ and choose $K > 4p/3$ in Lemma 4.2, increasing it if necessary. Fix the endpoint-chain accuracy $\eta > 0$ below a fixed small geometric constant. All these choices precede the large constant A below.

Let $R \geq AS$, where $S = \delta + \sqrt{\ell}$. On G_K we have $\Delta = o(S)$, uniformly as $\xi, \ell \to 0$, since

$$\frac{\xi L + \ell}{\delta + \sqrt{\ell}} \leq \xi^{1/3} L + \sqrt{\ell} \longrightarrow 0\,.$$

Suppose $R < M \leq 2R$ on this event, and take a point $z = (t, v) \in P$ attaining M. For sufficiently small ξ, (4.3) forces $0 < t < 1$ and $t, 1 - t \geq |v| = M$. Indeed, a projection outside the segment gives Euclidean excess at least M; if $0 \leq t \leq |v|$, the first leg alone gives excess at least $(\sqrt{2} - 1)|v|$, and similarly at the other endpoint. These alternatives contradict $3\Delta = o(R)$.

Choose the interval J from (6.1) containing t. Its scale satisfies $R \leq \tau$. On its cylinder the point z has Euclidean detour

$$|z| + |z - e_1| - 1 \geq c\, R^2/\tau\,. \tag{7.4}$$

Near zero, for example, $|z| - t = |v|^2/(|z| + t) \geq cR^2/\tau$; the central and reflected intervals have the same bound. Choose $\eta < c/2$ in Lemma 6.1. The geodesic triangle equality and Markov's inequality yield

$$\mathbb{P}[\text{such a point occurs in } \mathcal{C}(J, R), G_K] \leq C_q \left(\frac{\tau}{R^2}\right)^q \left(\xi^q R^{q/2} + B_q^q\right)\,,$$

where $B_q = \xi\sqrt{m_p + \omega} + \ell + \xi^2$. The sum of τ^q over (6.1) and their reflections is bounded. Thus, without a logarithmic loss from the endpoint scales,

$$\mathbb{P}[R < M \leq 2R, G_K] \leq C_p(\xi^{2p} R^{-3p} + B_{2p}^{2p} R^{-4p})\,. \tag{7.5}$$

On G_K the height tends uniformly to zero, so shells exceeding a fixed small radius are empty for sufficiently small ξ.

Use (7.2) and sum dyadic shells $R_j = 2^j AS$. The portion below AS costs $(AS)^p$. The complement of G_K costs at most $C_p \xi^{K/2} = o(\delta^p)$ by the uniform $2p$-th moment of M. Consequently

$$\begin{aligned} m_p^p &\leq (AS)^p + C_p \sum_{j \geq 0} \left(\xi^{2p} R_j^{-2p} + \xi^{2p} m_p^p R_j^{-3p} + S^{4p} R_j^{-3p}\right) + o(\delta^p) \\ &\leq C_p A^p S^p + C_p A^{-3p} m_p^p\,. \end{aligned}$$

Here $\xi^{2p} = \delta^{3p} \leq S^{3p}$. All numbers are finite before this estimate is used. Choose A so that $C_p A^{-3p} \leq 1/2$ and absorb to obtain (7.3). □

Lemma 7.3 (The negative part of the exponent)**.** *For sufficiently small ξ,*

$$\lambda \geq -C_d \xi^2\,. \tag{7.6}$$

Proof. Since $\mathbb{E}F = 0$, Jensen gives $\mathbb{E}\rho(0, e_1) \geq 1$. At either of the two deterministic half-segments, the exact similarity formula writes the distance as $2^{-a_\xi} e^X$, where $\mathbb{E}X = 0$ and the basic concentration estimate gives $\mathbb{E}|e^X - 1 - X| \leq C\xi^2$. The spherical average at radius $1/2$ has bounded variance, and no independence from the transformed metric is needed. Taking expectations in the triangle inequality through $e_1/2$ gives

$$1 \leq \mathbb{E}\rho(0, e_1) \leq 2^{1 - a_\xi}(1 + C\xi^2) = 2^\lambda (1 + C\xi^2)\,.$$

Taking logarithms proves (7.6). □

Lemma 7.4 (The midpoint comparison)**.** *Fix $p \geq 4(d+1)$. For every $\eta > 0$, there is $C_{p,\eta} < \infty$ such that, whenever $\lambda \geq 0$ and ξ is sufficiently small,*

$$\lambda \leq \eta m_p^2 + C_{p,\eta}\xi\sqrt{m_p} + C_{p,\eta}\delta^2 \,. \tag{7.7}$$

Proof. Step 1: Retaining the deterministic power. Fix $0 \leq \lambda \leq \lambda_0 < 1/8$ and write $a = 1 - \lambda$. For $z = (1/2, v)$ with $|v| \leq R$, use longitudinal approximants $t_n^{\pm} = 1/2 \pm \ell_n$, where $\ell_n = \kappa 2^{-n}$, and round v at mesh $R\ell_n^{\beta}$. These have all the counting, summability and Gaussian properties in Lemma 6.1. Their Euclidean loss for either anchor is at most $C\kappa^{2\beta-1}R^2$.

Write $r = |z|$, $r_0 = |z_0^+|$ and $e_n = |z_n^+ - z_{n+1}^+|$. The long chords lie in a fixed subinterval of $(0,1)$ if κ, R are small. On that interval the derivative of $r^a - r$ has absolute value at most $C\lambda$, so

$$\left|(r_0^a - r_0) - (r^a - r)\right| \leq C\lambda|r_0 - r| \leq C\lambda\kappa^{\beta} \,.$$

For the short edges, $e_n \leq C\ell_n^{\beta}$, and

$$0 \leq e_n^a - e_n \leq \lambda e_n^{1-\lambda}\log(1/e_n) \,.$$

For κ small the function on the right, without λ, is increasing in e_n throughout this range. Therefore

$$\sum_{n\geq 0}(e_n^a - e_n) \leq C\lambda\,\kappa^{\beta(1-\lambda_0)}(1 + |\log\kappa|) \,.$$

Combining these bounds with the Euclidean chain inequality gives

$$r_0^a - \sum_{n\geq 0} e_n^a \geq r^a - A(\kappa)R^2 - B(\kappa)\lambda \,, \tag{7.8}$$

where $A(\kappa), B(\kappa) \to 0$ as $\kappa \downarrow 0$, uniformly in R and $\lambda \leq \lambda_0$. The same statement holds for the chain from e_1. The transverse mesh at level zero is $R\kappa^{\beta}$; keeping a coarser mesh there would lose the arbitrarily small coefficient of R^2.

Step 2: The stochastic remainder. Repeat the proof of Lemma 6.1, using (4.10) for the metric errors and (7.8) for their deterministic parts. For every small prescribed $\varepsilon > 0$, choose κ so that $2A(\kappa) \leq \varepsilon$ and $2B(\kappa) \leq (\log 2)/4$. There is then a nonnegative random variable W_R with

$$\|W_R\|_{L^{2p}} \leq C_{p,\varepsilon}\left(\xi\sqrt{R} + \xi\sqrt{m_p + \omega} + \xi\lambda + \xi^2\right) \,, \tag{7.9}$$

such that, simultaneously for $z = (1/2, v)$, $|v| \leq R$,

$$\rho(0,z) + \rho(z,e_1) - \rho(0,e_1) \geq |z|^a + |z - e_1|^a - 1 - \varepsilon R^2 - \tfrac{1}{4}(\log 2)\lambda - W_R \,.$$

The deterministic term is at least $2^{\lambda} - 1 \geq (\log 2)\lambda$, because both chord lengths are at least $1/2$.

Step 3: A comparison on an event of positive probability. Take the deterministic radius $R = 4(m_p + \delta)$. It tends to zero by Lemma 4.2. Markov gives $\mathbb{P}[M \leq R] \geq 3/4$. Also $\mathbb{P}[W_R \leq 4\,\|W_R\|_{2p}] \geq 3/4$. Their intersection has positive probability; no independence is used. On the first event the geodesic has a point on $t = 1/2$ with transverse coordinate of magnitude at most R. The triangle equality at that point therefore gives the deterministic inequality

$$c\lambda \leq \varepsilon R^2 + C_{p,\varepsilon}\left(\xi\sqrt{R} + \xi\sqrt{m_p + \omega} + \xi\lambda + \xi^2\right) \,.$$

First choose ε so that its coefficient of m_p^2 after inserting $R = 4(m_p + \delta)$ is at most a prescribed small multiple of η. The terms involving R also contribute at most $C_{p,\eta}(\delta^2 + \xi\sqrt{m_p})$. For $\lambda \geq 0$, apply (7.1) with a sufficiently small coefficient, chosen after $C_{p,\varepsilon}$, to absorb the $\xi\sqrt{\lambda\log(e/\lambda)}$ term into the left side. The term $\xi\lambda$ is absorbed by decreasing the upper bound on ξ. Finally $\xi^{3/2}L + \xi^2 L = o(\delta^2)$. This proves (7.7). □

Proof of Theorem B. Fix $p_0 = 4(d+1)$ and write $m = m_{p_0}$. Proposition 7.2 gives, with a fixed constant C_0,

$$m^2 \le C_0(\delta^2 + |\lambda|)\,.$$

If $\lambda \le 0$, Lemma 7.3 implies $|\lambda| \le C\xi^2 \le C\delta^2$, and the conclusions follow. Suppose $\lambda > 0$. Choose η in Lemma 7.4 so that $C_0\eta \le 1/4$. Then

$$\lambda \le \tfrac{1}{4}\lambda + C\delta^2 + C_\eta \xi\sqrt{m} \le \tfrac{1}{4}\lambda + C_\eta\delta^2 + C_\eta\xi\lambda^{1/4}\,.$$

The last step used $\sqrt{m} \le C(\delta^{1/2} + \lambda^{1/4})$ and $\xi\delta^{1/2} = \delta^2$. Young's inequality bounds the last term by $\lambda/4 + C_\eta\xi^{4/3}$. Absorption gives $\lambda \le C_d\delta^2$ and then $m \le C_d\delta$.

This proves (1.11). For any prescribed positive moment order, apply Proposition 7.2 at a fixed order at least $4(d+1)$ and at least the prescribed order. The exponent bound gives $m_p \le C_{d,p}\delta$. Monotonicity of L^p norms covers smaller orders. All small-parameter thresholds may depend on the fixed moment order. □

7.1 Growth of the height moments

The preceding estimates close at each fixed moment order. Keeping track of the leading constants gives a cubic exponential upper tail. The parameter threshold may depend on the fixed order, so the resulting bound is asymptotic for each fixed tail level.

Proposition 7.5. *Under the standing strong axioms there is $C_d < \infty$ such that, for every fixed real $p \ge p_0 = 4(d+1)$,*

$$\limsup_{\xi\downarrow 0} \frac{\|M_\xi\|_{L^p}}{\xi^{2/3}} \le C_d p^{1/3}\,. \tag{7.10}$$

Consequently there are $c_d, C_d > 0$ such that for every fixed $A \ge 1$,

$$\limsup_{\xi\downarrow 0} \mathbb{P}[M_\xi > A\xi^{2/3}] \le C_d \exp(-c_d A^3)\,. \tag{7.11}$$

Proof. Put $\delta = \xi^{2/3}$ and $m_p = \|M_\xi\|_p$. Throughout this proof p is fixed before ξ tends to zero, whereas a constant denoted by C_d is independent of p. Constants in a quantity $o_p(1)$ may depend on p.

Step 1: The leading constant in the point expansion. The deterministic estimate (4.8) has a dimensional constant:

$$\|\nu_P - \mu_0\|_{\mathcal{H}_d^*}^2 \le C_d M + C_d\Delta(1 + \log\mathcal{A} + |\log\Delta|)\,,$$

on the Euclidean good event. The general bound on its complement is $C_d(U+1)$. Fix $q \ge 2$. The good event has $\Delta \le C_K(\xi\log(e/\xi) + \ell)$, where $\ell = |1 - a_\xi|$ and K may be chosen after q. The logarithms of the metric Hölder constant have every fixed moment, as does U. Hölder on the exceptional event and a sufficiently large fixed K therefore give

$$\Big\|\|\nu_P - \mu_0\|_{\mathcal{H}_d^*}^2\Big\|_{q/2} \le C_d m_{q/2} + C_{d,q}\omega, \qquad \omega = \xi\log(e/\xi)^2 + \ell\log(e/\ell)\,. \tag{7.12}$$

The coefficient of $m_{q/2}$ is independent of q: it comes directly from the pointwise term $C_d M$. All uses of Hölder and the exceptional-event estimate affect only the second term.

The Gaussian gradient inequality (2.6) has constant $C\sqrt{q}$. Since the centered logarithmic remainder Z has derivative $\xi(\nu_P - \mu_0)$, it follows that

$$\|Z\|_q \le C_d\sqrt{q}\,\xi\sqrt{m_{q/2}} + C_{d,q}\xi\sqrt{\omega}\,.$$

In the exact pair expansion (4.11), this remainder is multiplied only by the deterministic factor r^{a_ξ}. Its leading coefficient is therefore unchanged up to a dimensional constant when bounded by $r^{3/4}$; we may take $a_\xi \in [7/8, 9/8]$. The deterministic power error costs $C_d \ell r^{3/4}$. All other errors are $C_{d,q}(\xi\sqrt{\omega} + \xi^2 + \xi\ell)r^{3/4}$. The already proved exponent estimate $\ell \le C_d\delta^2$ implies

$$\xi\sqrt{\omega} + \xi^2 + \xi\ell = o(\delta^2)\,.$$

Thus, uniformly over deterministic x, y in the fixed compact ball and $r = |x - y|$,

$$\|\rho(x,y) - r - \xi I_{xy}\|_q \le r^{3/4}\left(C_d\sqrt{q}\,\xi\sqrt{m_{q/2}} + C_d\delta^2 + o_q(\delta^2)\right)\,. \tag{7.13}$$

The o_q term is uniform in the endpoints, by the fixed-exponent pair expansion.

Step 2: Constants in the endpoint chains. Fix the chain accuracy η once, less than half the dimensional Euclidean detour constant used in the height-shell argument. Its mesh parameter κ is then fixed independently of q. For $q \ge 2p_0$, the metric-error series is uniformly summable because

$$\sum_{n\ge 0} 2^{2dn/q}2^{-9n/16} \le \sum_{n\ge 0} 2^{2dn/(2p_0)}2^{-9n/16} < \infty\,.$$

The number of coarse nodes depends only on d, κ, and its $1/q$ power is uniformly bounded in this range. Applying the maximum inequality at the same order q, the endpoint-chain variable $U_{J,R}$ therefore satisfies

$$\|U_{J,R}\|_q \le C_d\sqrt{q}\,\xi\sqrt{m_{q/2}} + C_d\delta^2 + o_q(\delta^2)\,.$$

For the Gaussian chain variable, the maximum bound at level n is at most $C_d\sqrt{R}\,2^{-3n/8}\sqrt{q+n}$. Summing it, including the coarse level, gives

$$\|V_{J,R}\|_q \le C_d\sqrt{q}\sqrt{R}\,.$$

Both constants are uniform over the longitudinal blocks, including those near the endpoints. No moment larger than q was used for U.

Step 3: Shell summation. Set $q = 2p$ and absorb the fixed-p error into δ^2 by making ξ small. The geodesic witness and block sum from Proposition 7.2 now give, for $R \ge Bp^{1/3}\delta$ and a sufficiently large fixed B,

$$\mathbb{P}[R < M \le 2R, G] \le C_d^{2p}\left(p^p\xi^{2p}R^{-3p} + (p^p\xi^{2p}m_p^p + \delta^{4p})R^{-4p}\right)\,. \tag{7.14}$$

Here G is the Euclidean good event with an exceptional exponent chosen after p. The block factor $\sum_J \tau_J^{2p}$ is uniformly bounded for $p \ge p_0$. All factors such as the geometric detour constant, the L^{2p} triangle inequality, and the dyadic shell factor are absorbed in C_d^{2p}; no additional power of p is lost.

Let $R_0 = Bp^{1/3}\delta$. Multiply (7.14) by $(2R)^p$ and sum over $R = 2^jR_0$. The contribution of G^c is $o_p(\delta^p)$ by a fixed higher height moment and a sufficiently large exceptional exponent. The constants from these higher moments may depend on p; they multiply terms that vanish at fixed p, before the final moment optimization. The geometric sums yield

$$\begin{aligned} m_p^p &\le R_0^p + C_d^{2p}\left(p^p\xi^{2p}R_0^{-2p} + p^p\xi^{2p}m_p^pR_0^{-3p} + \delta^{4p}R_0^{-3p}\right) + o_p(\delta^p) \\ &\le B^pp^{p/3}\delta^p + C_d^{2p}B^{-2p}p^{p/3}\delta^p + (C_d^2B^{-3})^pm_p^p + C_d^{2p}B^{-3p}p^{-p}\delta^p + o_p(\delta^p)\,, \end{aligned}$$

where $\xi^{2p} = \delta^{3p}$. Choose B independently of p so that $(C_d^2B^{-3})^{p_0} < 1/2$. Absorb the height term and take the pth root followed by the upper limit as $\xi \downarrow 0$. This proves (7.10).

For large fixed A, choose the fixed order $p = (A/(eC_d))^3 \ge p_0$. Markov's inequality and (7.10) give an upper limit at most e^{-p}. Enlarging the prefactor covers the bounded remaining range $1 \le A < eC_dp_0^{1/3}$, proving (7.11). □

8 Path tightness

We prove the tightness and longitudinal assertions of Theorem C. This section uses the scalar upper estimates of Theorem B. The lower tail is needed only to exclude the zero path from the limiting laws, after the tightness proof.

Put $\delta = \xi^{2/3}$. Fix the exponent $\alpha = 3/4$. The deterministic-endpoint expansion and the completed height/exponent estimates give, for every prescribed fixed $q \geq 2$,

$$\|\rho(x,y) - |x-y| - \xi I_{xy}\|_{L^q} \leq C_q \delta^2 |x-y|^{3/4}, \qquad x, y \in \overline{B}_3 \,. \tag{8.1}$$

Indeed $\xi\sqrt{m_{q/2}} = O(\delta^2)$, $\ell = O(\delta^2)$, and $\xi\sqrt{\omega} + \xi^2 = o(\delta^2)$. All moment orders remain fixed before taking $\xi \downarrow 0$.

Lemma 8.1 (A triangle with three variable endpoints)**.** *Fix $K \geq 1$ and $q \geq 64(d+1)$. Let $0 < \tau \leq 1$, $s \in [0, 1-\tau]$, and*

$$\delta\sqrt{\tau} \leq R \leq \tau/100 \,.$$

Consider triples

$$a = (s,u), \quad z = (s + \tau/2, w), \quad b = (s+\tau, v) \,,$$

with $u, w, v \in \mathbb{R}^{d-1}$, all of magnitude at most $K\delta$, satisfying

$$|v-u| \leq \tau, \qquad \left| w - \frac{u+v}{2} \right| \leq 2R \,.$$

For every $\eta > 0$ and all sufficiently small ξ, depending on the fixed K, q, η, there are nonnegative random variables $U_{s,\tau,R}$ and $V_{s,\tau,R}$ such that, simultaneously over these triples,

$$\begin{aligned} \rho(a,z) + \rho(z,b) - \rho(a,b) &\geq |a-z| + |z-b| - |a-b| - \eta R^2/\tau - U_{s,\tau,R} - \xi V_{s,\tau,R}, \\ \|U_{s,\tau,R}\|_q &\leq C_{K,q,\eta} \delta^2 \tau^{3/4 - 4d/q}, \\ \|V_{s,\tau,R}\|_q &\leq C_{K,q,\eta} \tau^{-4d/q} (\sqrt{\tau R} + R\sqrt{\log(e/\tau)}) \,. \end{aligned}$$

Proof. We give moving-endpoint chains for each of the three distances. The side segments have longitudinal length $l = \tau/2$ and slope at most $1 + 4R/\tau < 2$; the long segment has longitudinal length $l = \tau$ and slope at most one.

For any one pair $a = (t,u), b = (t+l, v)$, write $k_n = \kappa 2^{-n}$, with κ small, and $\beta = 3/4$. Round u, v componentwise at mesh Rk_n^β, obtaining u_n, v_n, and set $a'_n = (t, u_n)$, $b'_n = (t+l, v_n)$. For $\varepsilon \in \{-1, 1\}$ define

$$a_n = a'_n + \varepsilon k_n (b'_n - a'_n), \qquad b_n = b'_n - \varepsilon k_n (b'_n - a'_n) \,.$$

The sign $\varepsilon = -1$ extends the endpoints outwards and is used for a lower bound on the distance; the sign $\varepsilon = 1$ moves them inwards and is used for an upper bound. The two side distances use the outward choice and the long distance the inward choice. All approximants converge to their targets, all long chord lengths are comparable to τ, and the short edges have length at most $C\tau k_n^\beta$.

Let $e = (b-a)/|b-a|$ be the direction of the exact unrounded chord. Each approximant is within CRk_n^β of its extension or contraction along this chord. For the outward chain, orient the polygon from a_n through a_{n+1}, b_{n+1} to b_n. For the inward chain, orient it from a_{n+1} through a_n, b_n to b_{n+1}. Its two short displacements have the form $t_j e + E_j$, with $t_j \asymp \tau k_n > 0$ and $|E_j| \leq CRk_n^\beta$; its long displacement has $t_j \asymp \tau$ and the same error bound. For arbitrary E,

$$0 \leq |te + E| - e \cdot (te + E) \leq \frac{|E|^2}{2t}, \qquad t > 0 \,.$$

Indeed, on writing $v = te + E$, the upper inequality is equivalent to $2t|v| \le |v|^2 + t^2$. It does not require a small short-edge slope. If d_j are the three oriented displacements,

$$\sum_j |d_j| - \left|\sum_j d_j\right| \le \sum_j \left(|d_j| - e \cdot d_j\right) \le C\frac{R^2}{\tau} k_n^{2\beta-1} .$$

The exact direction is used only to prove this bound. The actual approximants depend on rounded endpoints and form deterministic finite sets.

In the outward case the two nonnegative triangle excesses at level n add to

$$|a_{n+1} - b_{n+1}| + |a_n - a_{n+1}| + |b_n - b_{n+1}| - |a_n - b_n| .$$

In the inward case they add to

$$|a_n - b_n| + |a_n - a_{n+1}| + |b_n - b_{n+1}| - |a_{n+1} - b_{n+1}| .$$

Summing these identities bounds the total deterministic loss by $C(R^2/\tau)\kappa^{2\beta-1}$. Choose κ to make the combined loss for all three pairs at most $\eta R^2/\tau$.

The corresponding metric comparisons are

$$\rho(a, b) \ge \rho(a_0, b_0) - \sum_{n\ge 0} [\rho(a_n, a_{n+1}) + \rho(b_n, b_{n+1})] ,$$

for the outward chains, and the reversed inequality with a plus sign before the sum for the inward chains. They follow first for finite chains and then by continuity and summability.

We record the counts to preserve uniformity in ξ. A rounded endpoint has at most $C_d(1 + K\delta/R)^{d-1} k_n^{-\beta(d-1)}$ values. For edges or coupled triples we can use the conservative bound

$$N_n \le C_{K,\kappa,d}(1 + \delta/R)^{6(d-1)} 2^{6\beta(d-1)n} \le C_{K,\kappa,d}\tau^{-3(d-1)} 2^{(9/2)(d-1)n} .$$

Only values arising from admissible pairs or triples are counted. Thus all long-chord and altitude bounds continue to hold on the finite index sets. The L^q maximum bound by $N_n^{1/q}$, together with (8.1), makes the total coarse and short-edge metric error at most

$$C_{K,q,\kappa}\delta^2\tau^{3/4-4d/q}\left(1 + \sum_{n\ge 0} 2^{-9n/16+(9/2)(d-1)n/q}\right) .$$

The series converges for the chosen q. The basic distance moments give almost-sure convergence of the metric-edge sums as well.

For the Gaussian part, each triangle defect in the preceding telescoping identities has long chord comparable to τ and altitude at most CRk_n^β. The line-integral triangle estimate therefore bounds its L^2 norm by

$$C\sqrt{\tau R}\, k_n^{\beta/2} + CRk_n^\beta\sqrt{\log(e/\tau)} .$$

The two triangle identities are the same with Euclidean lengths replaced by line integrals; their short-edge terms telescope with exactly the signs in the metric comparisons. The count bound and Gaussian moments make the sum of their maximal absolute values bounded in L^q by

$$C_{K,q,\kappa}\tau^{-4d/q}\left(\sqrt{\tau R} + R\sqrt{\log(e/\tau)}\right) .$$

Here $\beta/2 = 3/8 > (9/2)(d-1)/q$.

It remains to bound $T(a,z,b) = I_{az} + I_{zb} - I_{ab}$ uniformly over the admissible triples. Round all three transverse coordinates at the same meshes, keeping their longitudinal coordinates fixed. The coarse variables have the preceding L^2 bound, since rounding increases the altitude only by $CR\kappa^\beta$. For adjacent levels, corresponding endpoints move by $D_n \le CRk_n^\beta$, while all segment lengths are comparable to τ. Transport of their normalized arclength measures and their mass difference give

$$\|I_{xy} - I_{x'y'}\|_2 \le C\sqrt{\tau D_n} + CD_n\sqrt{\log(e/\tau)}\,.$$

Indeed, translate to one endpoint, rescale by τ, and apply the logarithmic transport lemma to obtain energy $C\tau D_n$ for the equal-mass part; the length difference is at most $2D_n$. Apply this to the three segments, then use the same counts and summation. This gives the same L^q bound for $\sup|T|$. All constructions are first made on a deterministic countable dense set and then extended by continuity of the metric and Gaussian line-integral process. Combining the three distance comparisons proves the lemma. □

8.1 A modulus of continuity

Proof of the tightness assertions in Theorem C. Parametrize the geodesic at constant ρ-speed on $[0,1]$. Let $\Delta = \sup_{x,y\in\overline{B}_3} |\rho(x,y) - |x-y||$. With probability tending to one,

$$P_\xi \subset B_2, \qquad \Delta \le C\xi\log(e/\xi)\,.$$

This follows from the quantitative Euclidean estimate and the proved $\ell = O(\xi^{4/3})$. On this event, geodesicity and the constant-speed identity imply, for every t,

$$\big||P_\xi(t)| - t\big| \le 2\Delta, \qquad |P_\xi(t)| + |P_\xi(t) - e_1| - 1 \le 3\Delta\,.$$

Since $|P - e_1| \ge 1 - P_1$, these inequalities give

$$\sup_t |P_{\xi,1}(t) - t| \le 5\Delta = o(\delta)\,. \tag{8.2}$$

For $s \in (0,1)$ let t_s be the first time the path hits the plane with first coordinate s. Put $t_0 = 0$ and $t_1 = 1$; the latter is the terminal-time convention, even if the plane was hit earlier. These times are increasing in s, and (8.2) gives $|t_s - s| \le 5\Delta$. Write $v_s = P_\xi^\perp(t_s)$. On the event $M_\xi \le K\delta$, all $|v_s| \le K\delta$.

Set $\tau_* = \delta^{4/3}$ and let $N = N(\xi)$ satisfy $\tau_N = 2^{-N} \in [\tau_*, 2\tau_*)$. The two scale separations are $\Delta/\tau_* = O(\xi^{1/9}\log(e/\xi)) \to 0$ and $\tau_*/\delta = \xi^{2/9} \to 0$. For each dyadic interval $[s, s+\tau]$ of length $\tau = 2^{-n} \ge \tau_N$, consider the successive geodesic points at longitudinal positions $s, s+\tau/2, s+\tau$. Their chord lengths are at most their longitudinal separation plus $C\Delta$, because their metric distance is at most $(1+\Delta)$ times the difference of the corresponding t_s, and the additive Euclidean error is Δ. Consequently,

$$|v_{s+\tau} - v_s| \le C\sqrt{\tau\Delta + \Delta^2}, \qquad W_{s,\tau} := \left|v_{s+\tau/2} - \frac{v_s + v_{s+\tau}}{2}\right| \le C\sqrt{\tau\Delta + \Delta^2}\,.$$

Since $\Delta/\tau_* \to 0$, these triples have slope at most one and $W_{s,\tau} \le \tau/100$ for small enough ξ, uniformly over the indicated scales.

If $R < W_{s,\tau} \le 2R$ and $R \ge \delta\sqrt{\tau}$, their Euclidean triangle excess is at least cR^2/τ. This follows either by uniform convexity of the Euclidean norm with bounded slopes, or by the Hessian of

$u \mapsto \sqrt{(\tau/2)^2 + |u|^2}$ on the ball of slopes at most two. Choose $\eta < c/2$ in Lemma 8.1. The metric triangle excess is zero by geodesicity. Markov's inequality gives

$$\mathbb{P}[R < W_{s,\tau} \le 2R,\ G,\ M_\xi \le K\delta] \le C_{K,q}(\delta^{2q}\tau^{7q/4-4d}R^{-2q} + \xi^q\tau^{3q/2-4d}R^{-3q/2} + \xi^q\tau^{q-4d}R^{-q}\log(e/\tau)^{q/2})\,.$$

Here G is the preceding high-probability Euclidean event. Empty shells above $\tau/100$ are discarded. Summing shells from $R_0 = \delta\sqrt{\tau}$ and using $\xi\delta^{-3/2} = 1$ gives

$$\mathbb{P}[W_{s,\tau} > \delta\sqrt{\tau},\ G,\ M_\xi \le K\delta] \le C_{K,q}\left(\tau^{3q/4-4d} + \delta^{q/2}\tau^{q/2-4d}\log(e/\tau)^{q/2}\right) \le C_{K,q}\tau^{q/4-4d}\,.$$

The condition $\delta\sqrt{\tau} \le \tau/100$ holds uniformly for $\tau \ge \tau_*$ when ξ is sufficiently small. There are τ^{-1} cells at scale τ. Thus, with $\kappa = q/4 - 4d - 1 > 0$, for every fixed n_0 and small enough ξ,

$$\mathbb{P}[\exists n_0 \le n < N,\, \exists s:\ W_{s,2^{-n}} > \delta 2^{-n/2},\ G,\ M_\xi \le K\delta] \le C_{K,q}2^{-\kappa n_0}\,.$$

Let V_n be the piecewise affine interpolation of v_s on the dyadic grid of mesh 2^{-n}. On the complementary event,

$$\|V_{n+1} - V_n\|_\infty \le \delta 2^{-n/2}, \qquad n_0 \le n < N\,,$$

so $\|V_N - V_{n_0}\|_\infty \le C\delta 2^{-n_0/2}$. Meanwhile V_{n_0}/δ has norm at most K and Lipschitz constant at most $2K2^{n_0}$.

Finally,

$$\|P_\xi^\perp - V_N\|_\infty \le C(\tau_N + \Delta) = o(\delta)\,.$$

To see this, if $t \in [s, s+\tau_N]$, then $|t - t_s| \le \tau_N + 5\Delta$. The constant-speed identity $\rho(P_\xi(t), P_\xi(t_s)) = |t - t_s|\rho(0, e_1)$ holds for either order of the two times. Geodesicity and the additive Euclidean approximation therefore bound $|P_\xi(t) - P_\xi(t_s)|$ by $C(\tau_N + \Delta)$. The same bound controls the difference of the two adjacent grid values and hence the affine interpolation.

For any desired probability error, first choose K from scalar height tightness, then choose n_0 large enough to make both $C2^{-n_0/2}$ and $C_{K,q}2^{-\kappa n_0}$ small. The bounded Lipschitz family V_{n_0}/δ has a uniform modulus of continuity. The displayed approximation, together with (8.2), proves

$$\lim_{h\downarrow 0}\limsup_{\xi\downarrow 0}\mathbb{P}\left[\sup_{|t-u|\le h}|X_\xi(t) - X_\xi(u)| > \varepsilon\right] = 0, \qquad \varepsilon > 0\,. \tag{8.3}$$

The sup norms are bounded in probability and $X_\xi(0) = 0$. This is the tightness criterion along every sequence $\xi_n \to 0$ in the uniform topology. The first coordinate converges uniformly to zero in probability by (8.2). □

9 The lower tail

The local coordinate-change identity (1.3) is imported from [C, Theorem 1.1, p. 2] for $d \ge 3$ and [GMc, Theorem 1.1] for $d = 2$. Its domain, perturbation and fixed-map hypotheses are exactly those used below. We bend a thin tube in $d-1$ transverse directions and compare the resulting geodesic events under one Gaussian law. A bound for the expected supremum of the Cameron–Martin likelihoods allows all the disjoint events to be counted together.

Proposition 9.1. *Fix $d \geq 2$ and a parameter for which the strong axioms hold and the 0-to-e_1 geodesic exists and is almost surely unique. Put $Q = Q_\xi \geq 1$. There is a constant $C_d < \infty$, independent of this parameter, such that for every $r > 0$,*

$$\mathbb{P}[M \leq r] \leq C_d (Q^{2/3} r)^{d-1} . \tag{9.1}$$

In particular, if $Q = d/\gamma + \gamma/2$, then, uniformly for sufficiently small γ and all $b > 0$,

$$\mathbb{P}[M \leq b\gamma^{2/3}] \leq C_d b^{d-1} .$$

As usual, the assertion is vacuous when its right side is larger than one.

Proof. Write $m = d - 1$ and $x = (s, y) \in \mathbb{R} \times \mathbb{R}^m$. We use the same gauge throughout: the field is pinned by its average against the uniform probability measure σ_d on S^{d-1}. Cameron–Martin functions are represented in this gauge. If g is smooth and compactly supported, its pinned representative is

$$f = g - \langle g, \sigma_d \rangle ,$$

and its Cameron–Martin norm is

$$\|f\|_{\mathcal{H}_d}^2 = \mathfrak{c}_d \int_{\mathbb{R}^d} |\zeta|^d |\widehat{g}(\zeta)|^2 \, d\zeta, \qquad \mathfrak{c}_d > 0 . \tag{9.2}$$

Subtracting a constant does not change this norm. It also does not change geodesic images, by Weyl scaling.

Step 1: A family of bent tubes. For $u \in \mathbb{R}^m$ small, let ϕ_u be the endpoint-fixing Möbius bend whose midpoint image is $(1/2, u)$. More explicitly, for $u \neq 0$ set

$$v = u/|u|, \qquad \theta = 2 \arctan(2|u|), \qquad a_\theta = (\cos\theta - 1)e_1 - \sin\theta \, v ,$$

and, with O_θ the rotation by θ in the (e_1, v) plane, set

$$\phi_u(x) = O_\theta \frac{x + a_\theta |x|^2}{1 + 2a_\theta \cdot x + |a_\theta|^2 |x|^2} .$$

For $u = 0$ use the identity. The inverse is denoted by ψ_u. Both maps fix $0, e_1$ and preserve the plane $H = \{s = 1/2\}$. The identities

$$(\phi_u(x))_1 - \tfrac{1}{2} = \frac{s - 1/2}{1 + 2a_\theta \cdot x + |a_\theta|^2 |x|^2}, \qquad \phi_u(e_1/2) = (1/2, u) ,$$

verify these assertions directly. Their poles are outside a fixed ball containing the unit sphere and all domains below, provided $|u|$ is small enough.

Let $T_r = \{x : \operatorname{dist}(x, [0, e_1]) \leq r\}$. There are fixed $K_d, R_d > 0$ such that, whenever $0 < R \leq R_d$, $|u| \leq R$ and $0 < r \leq R/100$,

$$\phi_u(T_r) \subset U_R := \{-1/4 < s < 5/4, \ |y| < K_d R\}, \qquad \phi_u(T_r) \cap H \subset B_H((1/2, u), 2r) . \tag{9.3}$$

Indeed the maps and their inverses are uniformly close to the identity on the relevant fixed ball, and their Lipschitz constants on these tubes are at most two after reducing R_d. Preservation of H gives $\phi_u(T_r) \cap H = \phi_u(T_r \cap H)$, which proves the second inclusion.

Step 2: The Cameron–Martin cost. Choose fixed smooth cutoffs $\eta(s)$ and $\chi(z)$, equal to one on $[-1/4, 5/4]$ and $\overline{B}_{K_d}$ respectively, and supported in $[-1/2, 3/2]$ and B_{2K_d} respectively. Let σ_{ψ_u} be the conformal length scale of ψ_u. Define

$$g_{b,R}(s,y) = Q\eta(s)\chi(y/R)\log\sigma_{\psi_{Rb}}(s,y), \qquad |b| \le 1\,,$$

extended by zero off the cutoff support, and let $f_{b,R} = g_{b,R} - \langle g_{b,R}, \sigma_d\rangle$. The cutoffs avoid every pole. On U_R, the function $g_{b,R}$ is exactly the conformal coordinate drift.

The dependence on b is smooth, including at $b = 0$. To check this and the required bounds explicitly, the inverse conformal factor is

$$\sigma_{\psi_u}(s,y)^{-1} = 1 + \frac{8u\cdot y + 16|u|^2(s^2 - s + |y|^2)}{1 + 4|u|^2}\,.$$

Consequently,

$$\sigma_{\psi_{Rb}}(s,Rz)^{-1} = 1 + R^2\frac{8b\cdot z + 16|b|^2(s^2 - s + R^2|z|^2)}{1 + 4R^2|b|^2}\,.$$

It follows that

$$g_{b,R}(s,y) = QR^2 G_{b,R}(s, y/R)\,,$$

where $G_{b,R}$ and each of its derivatives in b have one fixed compact support and uniformly bounded smooth norms. The quotient $R^{-2}\log(1 + R^2 w)$ extends smoothly to $R = 0$, which makes the uniformity transparent.

Applying (9.2) and changing transverse Fourier variables gives

$$\|f_{b,R}\|_{\mathcal{H}_d}^2 = \mathfrak{c}_d Q^2 R^3 \int_{\mathbb{R}\times\mathbb{R}^m} (R^2\tau^2 + |\zeta|^2)^{d/2} |\widehat{G}_{b,R}(\tau,\zeta)|^2\, d\tau\, d\zeta\,.$$

The same identity applies after differentiation in b. Uniform smooth norms therefore imply

$$\sup_{|b|\le 1} \|f_{b,R}\|_{\mathcal{H}_d} \le C_d Q R^{3/2}, \qquad \|f_{b,R} - f_{b',R}\|_{\mathcal{H}_d} \le C_d Q R^{3/2}|b - b'|\,. \tag{9.4}$$

This argument applies without alteration in odd dimensions.

Fix a sufficiently small $c > 0$, depending only on d, and set

$$R = cQ^{-2/3}\,.$$

For $Q \ge 1$ we have $R \le R_d$, and the bounds in (9.4) are uniform in Q.

Step 3: A common family of likelihoods. Let W be the isonormal Gaussian process on the pinned Cameron–Martin space. Define

$$\mathcal{L}_b = \exp\left(W(f_{b,R}) - \tfrac{1}{2}\|f_{b,R}\|_{\mathcal{H}_d}^2\right)\,.$$

The Cameron–Martin formula has the orientation

$$\mathbb{P}[h + f_{b,R} \in E] = \mathbb{E}\left[\mathbf{1}_{\{h\in E\}}\mathcal{L}_b\right]\,, \tag{9.5}$$

for every measurable field event E. All likelihoods in this formula are defined on the same unshifted Gaussian probability space.

By (9.4), the centered Gaussian process $Z_b = W(f_{b,R})$ on the fixed unit ball of $\mathbb{R}^m$ has uniformly bounded variance and canonical distance at most $C_d|b-b'|$. It has a continuous version, and

$$\mathbb{E}\sup_{|b|\le 1}\mathcal{L}_b \le C_d\,. \tag{9.6}$$

For completeness, dyadic nets in the parameter ball have at most $C_d 2^{mn}$ points at level n. The maximum Gaussian increment from one level to the next has L^p norm at most $C_d 2^{-n}\sqrt{p+n}$, by the elementary Gaussian maximum estimate. Summing the levels gives $\|\sup_b|Z_b|\|_p \le C_d\sqrt{p}$ for $p \ge 1$. Expanding the exponential and using $p! \ge (p/e)^p$ shows $\mathbb{E}\exp(\sup_b|Z_b|) \le C_d$, which proves (9.6).

Step 4: Counting disjoint geodesic events. Suppose first that $r \le R/100$. Choose a deterministic maximal $8r$-separated family $u_1,\dots,u_N$ in $\overline{B}_R \subset \mathbb{R}^m$. Its $8r$-balls cover $\overline{B}_R$, so

$$N \ge (R/(8r))^m\,. \tag{9.7}$$

Write $b_k = u_k/R$, and let B_k be the event that the internal metric on the *same* domain U_R has a 0-to-e_1 minimizing curve whose image is contained in $\phi_{u_k}(T_r)$. By locality this is an event of the restricted field. Existence can be expressed by projecting the Borel graph of constant-speed minimizing curves in $C([0,1],\phi_{u_k}(T_r))$; the constant-speed identities are checked at rational times. Thus the event is analytic and universally measurable. The equivalent Gaussian laws used below have the same null sets, so the Cameron–Martin formula applies in their common completion.

By Lemma 4.1, the internal minimizing curve in U_R is almost surely unique whenever it exists. Consequently the events B_k are pairwise disjoint almost surely: a continuous curve from 0 to e_1 must cross H, while the relevant cross sections in (9.3) are disjoint. If two events held on the full-probability uniqueness event, their internal minimizers would be the same curve, which is impossible.

Put $A = \{M \le r\}$ and $p = \mathbb{P}(A)$. On A, the global geodesic lies in $T_r \subset \psi_{u_k}(U_R)$. Its conformal image is the unique internal geodesic in U_R for the transformed field. Uniqueness follows because any competing internal minimizer would pull back to another global minimizer of the original metric.

To identify the transformed field law, set locally

$$\widetilde{h}_k = h\circ\psi_{u_k} - \langle h\circ\psi_{u_k},\sigma_d\rangle\,.$$

All maps are defined on a fixed ball containing the unit sphere and U_R. The identity

$$|\psi_{u_k}(x)-\psi_{u_k}(y)|^2 = \sigma_{\psi_{u_k}}(x)\sigma_{\psi_{u_k}}(y)|x-y|^2\,,$$

shows that the pullback logarithmic covariance differs from the original one by a sum of one-variable terms. Subtracting the average against the same σ_d removes these terms. Thus $\widetilde{h}_k$ has exactly the canonical pinned Gaussian law on that fixed ball. No independence from the event A is asserted or required.

Conformal covariance, locality, the equality of $g_{b_k,R}$ with the coordinate drift on U_R, and invariance of geodesic images under constant shifts give

$$A(h) \implies B_k(\widetilde{h}_k + g_{b_k,R}), \qquad p \le \mathbb{P}[B_k(h+f_{b_k,R})]\,.$$

Using (9.5), summing over k, and using disjointness on the unshifted probability space, we obtain

$$Np \le \sum_{k=1}^{N}\mathbb{E}[\mathbf{1}_{B_k(h)}\mathcal{L}_{b_k}] \le \mathbb{E}\sup_{|b|\le 1}\mathcal{L}_b \le C_d\,.$$

Equation (9.7) and $R = cQ^{-2/3}$ now give (9.1). If $r > R/100$, increase C_d so that the right side of (9.1) is at least one. This handles every $r > 0$. Finally, $Q^{2/3}\gamma^{2/3} = (d + \gamma^2/2)^{2/3}$ is bounded for small γ, proving the last assertion. □

9.1 A lower bound for the upper tail

A single bent tube gives a complementary estimate. In contrast to the packing argument, its radius may be any sufficiently small number.

Proposition 9.2. *Fix a parameter at which the strong axioms hold and the global geodesic from* 0 *to* e_1 *exists and is almost surely unique. There are dimensional constants* $c_d, C_d > 0$*, independent of the parameter, such that*

$$\mathbb{P}[M > r] \geq \frac{1}{4}\exp(-C_d Q^2 r^3), \qquad 0 < r < c_d\,. \tag{9.8}$$

Consequently, for every fixed $A \geq 1$ *and all sufficiently small* ξ*,*

$$\mathbb{P}[M_\xi > A\xi^{2/3}] \geq \frac{1}{4}\exp(-C_d A^3)\,. \tag{9.9}$$

Every weak subsequential limit of $M_\xi/\xi^{2/3}$ *has unbounded support.*

Proof. Use the notation from the packing proof, with $R = 100r$ and one transverse vector u satisfying $|u| = R$. Choose c_d so that $R \leq R_d$. Set

$$A_r = \{M \leq r\}, \qquad p = \mathbb{P}(A_r)\,.$$

Let B_r be the event that the internal metric in the same deterministic domain U_R has a minimizing path from 0 to e_1 whose image lies in $\phi_u(T_r)$. This is a universally measurable local event, by the argument in the preceding proposition. The two tubes lie in U_R, and their sections in $H = \{x_1 = 1/2\}$ lie respectively in $B_H((1/2, 0), r)$ and $B_H((1/2, u), 2r)$. These sections are disjoint.

On A_r, the global minimizer lies in $T_r \subset U_R$, so the internal and global endpoint distances agree. If B_r also occurred, its internal minimizer would therefore be a global minimizer. Global uniqueness would make the two paths coincide. Their common path would lie in both tubes and would have to cross H, which is impossible. Hence

$$\mathbb{P}(B_r) \leq 1 - p\,.$$

Write $f = f_{u/R,R}$ for the pinned shift of the preceding proof. It equals $Q \log \sigma_{\psi_u}$ on U_R up to a constant. The estimate (9.4), before the choice $R = cQ^{-2/3}$, gives

$$\|f\|_{\mathcal{H}_d}^2 \leq C_d Q^2 R^3 \leq C_d' Q^2 r^3\,,$$

where the last constant includes the fixed factor 100^3. On A_r, the image of the original path under ϕ_u is minimizing for the internal metric

$$(x, y) \longmapsto D_h(\psi_u x, \psi_u y; \psi_u U_R) = D_{h\circ\psi_u + Q\log\sigma_{\psi_u}}(x, y; U_R)\,,$$

and lies in $\phi_u(T_r)$. Normalize $h \circ \psi_u$ by the same unit-sphere average. Its local Gaussian law is the canonical pinned law, as in the packing proof. Locality and constant-shift invariance thus give

$$p \leq \mathbb{P}[B_r(h + f)] = \mathbb{E}[\mathbf{1}_{B_r(h)} L_f], \qquad L_f = \exp(W(f) - \|f\|_{\mathcal{H}_d}^2/2)\,.$$

Only B_r is evaluated under the transformed local law. The disjointness with A_r holds on the original global field probability space. Since $\mathbb{E}L_f^2 = \exp(\|f\|_{\mathcal{H}_d}^2)$, Cauchy–Schwarz yields

$$p^2 \le (1-p)\exp(C_d Q^2 r^3)\,.$$

For $p \ge 1/2$ this proves (9.8); for $p < 1/2$ the conclusion is immediate. Since $Q_\xi \xi \to 1$, the scaled estimate follows for each fixed A once $A\xi^{2/3} < c_d$.

If $M_{\xi_n}/\xi_n^{2/3}$ converges in law to Y, the closed-set Portmanteau inequality gives, for every fixed $A \ge 1$,

$$\mathbb{P}[Y \ge A] \ge \limsup_n \mathbb{P}[M_{\xi_n}/\xi_n^{2/3} \ge A] \ge \frac{1}{4} e^{-C_d A^3} > 0\,.$$

This proves the support assertion. □

10 Completion of the main results

Proof of Theorem A. Existence and uniqueness follow from Sections 3 and 4. Theorem B gives every fixed positive moment bound (1.6). The imported coordinate identity (1.3), with the scope recorded in Section 1, supplies the conformal input to Proposition 9.1. Since $Q_\xi \xi = a_\xi \to 1$, that proposition gives

$$\mathbb{P}[M_\xi \le a\xi^{2/3}] \le C_d\big(a(Q_\xi\xi)^{2/3}\big)^{d-1} \le C_d a^{d-1}\,,$$

which is (1.7).

Choose $a > 0$ so that $C_d a^{d-1} < \varepsilon/2$. The first positive moment and Markov's inequality allow $A < \infty$ with $\mathbb{P}[M_\xi > A\xi^{2/3}] < \varepsilon/2$ uniformly for sufficiently small ξ. This proves (1.8). For $X = M_\xi/\xi^{2/3}$ and $0 < q < d-1$, layer cake gives

$$\mathbb{E}X^{-q} \le 1 + q\int_0^1 t^{-q-1}\mathbb{P}[X < t]\,\mathrm{d}t \le 1 + C_d q \int_0^1 t^{d-q-2}\,\mathrm{d}t < \infty\,,$$

uniformly in sufficiently small ξ. Propositions 7.5 and 9.2 give (1.9). If $M_{\xi_n}/\xi_n^{2/3}$ converges in law to Y, the open-set Portmanteau inequality, followed by a right limit in a, gives the small-ball estimate in (1.10). The same open-set inequality gives its upper-tail upper bound. For the lower bound, apply the closed-set inequality to $[A+\varepsilon, \infty)$ and let $\varepsilon \downarrow 0$. This gives

$$\mathbb{P}[Y > A] \ge \frac{1}{4} e^{-C_d A^3}, \qquad A \ge 1\,.$$

In particular, the support of Y is unbounded. Finally (1.5) transfers all assertions to the conventional γ-parameterization. □

Remaining assertions of Theorem C. The tightness and longitudinal assertions were proved in Section 8. On the Euclidean good event, (8.2) bounds every longitudinal overshoot beyond $[0,1]$ by 5Δ. Thus

$$0 \le \frac{M_\xi}{\delta} - \left\| \frac{P_\xi^\perp}{\delta} \right\|_\infty \le \frac{5\Delta}{\delta} \longrightarrow 0 \quad \text{in probability}\,.$$

If X_{ξ_n} converges in law to X, its endpoints and longitudinal coordinate therefore have the stated values, and M_{ξ_n}/δ_n converges in law to $\|X\|_\infty$. For $0 < a < a + \varepsilon < 1$, the open-set Portmanteau inequality gives

$$\mathbb{P}[\|X\|_\infty \le a] \le \mathbb{P}[\|X\|_\infty < a+\varepsilon] \le \liminf_n \mathbb{P}[M_{\xi_n}/\delta_n < a+\varepsilon] \le C_d(a+\varepsilon)^{d-1}\,.$$

Let $\varepsilon \downarrow 0$ and take $C_d \ge 1$ for the case $a = 1$. Finally let $a \downarrow 0$ to exclude the zero path. The norm $\|X\|_\infty$ also satisfies the upper-tail bounds in (1.10). □

References

[U] *Higher-dimensional exponential metrics: a proposed axiomatic existence and uniqueness proof.* AI-assisted research manuscript, revised 18 October 2026, 30 pp. Version supplied as `higher-dimensional-lqg-metric-revised.pdf`. The imported results are Definition 1.1 and Theorem 1.2 (p. 2), Theorem 1.4 (p. 3), Lemma 10.1 (pp. 26–27), Corollary 10.2 (p. 27), and Corollary 10.4 (p. 28). The manuscript labels its characterization and uniqueness argument as proposed research.

[C] *Conformal covariance of higher-dimensional exponential metrics: an axiomatic implication.* AI-assisted research note, 19 October 2026, 8 pp. Version supplied as `higher_dimensional_conformal_covariance.pdf`. We use Theorem 1.1 and Lemma 1.2 (p. 2), conditional on their stated metric inputs.

[CHZ] A. A. Contreras Hip and Z. Zhuang. *Weak exponential metrics for high-dimensional log-correlated Gaussian fields.* arXiv:2512.06292v1 (2025). https://arxiv.org/abs/2512.06292v1. The fixed-parameter moment inputs are Theorem 1.4(1)–(2) (p. 5), Proposition 4.7 (p. 38), and Proposition 4.11 (p. 40).

[DGZ] J. Ding, E. Gwynne, and Z. Zhuang. *Tightness of exponential metrics for log-correlated Gaussian fields in arbitrary dimension.* arXiv:2310.03996v5 (2026). https://arxiv.org/abs/2310.03996v5. We use the coarse exponent bounds and continuity in Proposition 1.1 (p. 4).

[GM] E. Gwynne and J. Miller. *Existence and uniqueness of the Liouville quantum gravity metric for* $\gamma \in (0, 2)$. Inventiones Mathematicae **223** (2021), 213–333. arXiv:1905.00383. https://arxiv.org/abs/1905.00383.

[GMc] E. Gwynne and J. Miller. *Conformal covariance of the Liouville quantum gravity metric for* $\gamma \in (0, 2)$. Annales de l'Institut Henri Poincaré, Probabilités et Statistiques **57** (2021), 1016–1031. arXiv:1905.00384. https://arxiv.org/abs/1905.00384.

[P] *Small-parameter LQG geodesics: deviation and functional tightness.* AI-assisted research manuscript, 2026, 16 pp.; repository version accessed 20 October 2026. https://github.com/leobon12/ai-generated-mathematics/blob/main/papers/small-parameter-lqg-geodesics.pdf.

[F] *First-order fluctuations of small-parameter exponential metrics.* AI-assisted research manuscript, 19 October 2026. Version supplied as `unified-first-order-exponential-metrics.pdf`. Sections 2–3 (pp. 4–9) provide the Gaussian calculus and entropy method retained in quantitative form here.